Distance Seidel Energy under Edge Deletion in Complete Multipartite Graphs

Guangfu Wang[1] and Xiaoling Zhang*[1]

[1]School of Mathematics and Information Sciences, Yantai University, Yantai, Shandong Province 264005, China, E-mail addresses: gfwang@ytu.edu.cn (G. Wang), zhangxling04@aliyun.com (X. Zhang)

Abstract

Let $G = K_{n_1,\dots,n_q}$ be a complete multipartite graph and let e be an edge such that $G - e$ is connected. We prove that deleting e strictly increases the distance Seidel energy:

$$E_{DS}(G - e) > E_{DS}(G).$$

This settles affirmatively a problem of Haritha and Chithra for complete bipartite graphs and extends it to every complete multipartite graph, including parts of size one. We also show that the single-edge hypothesis is essential: deleting a three-edge matching from $K_{1,3,3}$ decreases the distance Seidel energy.

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1 Introduction

The energy of a graph, introduced by Gutman [6], is the trace norm of its adjacency matrix. The effect of deleting an edge on graph energy subsequently developed into a separate line of study; foundational results and an account of the early questions can be found in [5, 1, 19]. Similar questions have since been asked for several matrices associated with a graph.

Let G be a finite simple connected graph. Its distance matrix $D(G)$ is indexed by $V(G)$ and has (u, v)-entry $d_G(u, v)$. We refer to the survey of Aouchiche and Hansen [2] for the history and spectral theory of distance matrices. The distance energy, introduced by Indulal, Gutman, and Vijayakumar [10], is the trace norm of $D(G)$. For this energy, Varghese, So, and Vijayakumar proved that deleting an edge from a complete bipartite graph increases the energy [18]. Tian, Li, and Cui obtained multipartite extensions [15], and Sun and Das completed the corresponding complete multipartite conjecture [13]. A correction to the proof appeared in [14].

The Seidel matrix $J - I - 2A(G)$ goes back to van Lint and Seidel [17]. Haemers introduced its energy in [7]. The behavior of Seidel energy under edge deletion has been studied for tripartite and 5-partite Turán graphs and, more recently, for independent-set blow-ups [16, 11, 12].

Haritha and Chithra [8] combined the distance and Seidel constructions by defining the *distance Seidel matrix*

$$D^S(G) = J - I - 2D(G).$$

*Corresponding author.

If its eigenvalues are $\partial_1^S(G), \dots, \partial_n^S(G)$, the *distance Seidel energy* is

$$E_{DS}(G) = \sum_{i=1}^{n} |\partial_i^S(G)|.$$

Haritha and Chithra determined the distance Seidel spectrum of a complete multipartite graph, including the eigenvalue 3 with multiplicity $\sum_i n_i - q$ [8, Theorems 3.7 and 3.8]. They also proved

$$E_{DS}(K_{a,b}) = 6(a + b - 2)$$

and established the desired strict inequality after edge deletion for $K_{2,n}$ and $K_{3,n}$, and asked whether it holds for all complete bipartite graphs. We reproduce their question in its original form.

Problem 1.1. *Let $a, b > 1$ and let e be an edge of $K_{a,b}$. Is*

$$E_{DS}(K_{a,b} - e) > E_{DS}(K_{a,b})$$

always true?

This is Problem 5.1 of Haritha and Chithra [8, Problem 5.1]. Our main theorem answers Problem 1.1 and proves the stronger statement suggested by the multipartite quotient structure.

Theorem 1.2. *Let $q \geq 2$ and $n_1, \dots, n_q \geq 1$, and put $G = K_{n_1,\dots,n_q}$. For every edge $e \in E(G)$ such that $G - e$ is connected,*

$$E_{DS}(G - e) > E_{DS}(G).$$

The restriction to one deleted edge is essential. The following exact counterexample, proved in Section 10, rules out the corresponding statement for matchings.

Proposition 1.3. *Let M be a perfect matching between the two parts of size 3 in $K_{1,3,3}$. Then*

$$E_{DS}(K_{1,3,3} - M) < E_{DS}(K_{1,3,3}).$$

We record the connectivity condition in Theorem 1.2.

Remark 1.4. The connectivity hypothesis has a simple form. If $q \geq 3$, then $G - e$ is automatically connected: the endpoints of e have a common neighbour in any third part. If $q = 2$ and the endpoint parts have sizes a and b, then $K_{a,b} - e$ is connected if and only if $a, b \geq 2$. Indeed, a singleton endpoint part makes the opposite endpoint isolated after deletion, whereas for $a, b \geq 2$ every endpoint remains joined to the rest of the graph by a path of length two or three.

The distinction between energy and spectral radius is already visible in a small example.

Remark 1.5. The theorem is not a formal consequence of spectral-radius monotonicity. For example,

$$\operatorname{Spec}\big(D^S(K_{2,2})\big) = \{3, 3, -1, -5\},$$

so $E_{DS}(K_{2,2}) = 12$, whereas twice the distance Seidel spectral radius is only 10. Thus control of a single extremal eigenvalue does not determine the energy.

The strict inequality is nontrivial. Although deleting an edge strictly increases the distance Seidel spectral radius [8, Section 4], the perturbation has both a positive and a negative direction, and the trace norm is not monotone under such an indefinite perturbation; see also Remark 1.5. Our proof uses three ingredients.

The multipartition first yields an orthogonal decomposition into contrast spaces and a reduced quotient block; the latter has at most one positive eigenvalue. Ky Fan's variational principle then settles all but one asymmetric configuration. In the remaining case, a compression of the third additive compound reduces the comparison to explicit inequalities in the part sizes. The longer polynomial identities are included in the appendices, and Supplementary File S1 gives the intermediate calculations needed to check them line by line.

For reference, the exhaustive case split is summarized in Table 1.

Table 1: Exhaustive proof roadmap for Theorem 1.2.

Configuration of the deleted edge	Result used
$q = 2$	Proposition 4.1
$q \geq 3$ and both endpoints are singleton parts	Proposition 5.1
$q \geq 3$ and exactly one endpoint is a singleton part	Proposition 5.2
$q \geq 3$, both endpoints are non-singleton, and $R \preceq 0$	Proposition 5.3
$q \geq 3$, R has a positive eigenvalue, and exactly two parts are non-singleton	Proposition 6.1
$q \geq 3$, R has a positive eigenvalue, at least three parts are non-singleton, and the endpoint parts have equal size	Proposition 7.1
$q \geq 3$, R has a positive eigenvalue, at least three parts are non-singleton, and the endpoint parts have unequal sizes	Section 8

The paper is organized as follows. Section 2 collects the matrix facts used throughout. Section 3 gives the multipartite orthogonal decomposition. The complete bipartite case is settled in Section 4. Edges involving singleton parts and the case of a negative-semidefinite reduced quotient are treated in Section 5. The case with only two non-singleton parts is handled in Section 6, and equal endpoint parts in an arbitrary multipartite graph are treated in Section 7. Unequal endpoint parts are treated in Section 8, and Section 10 proves Proposition 1.3. The appendices contain the algebraic reductions and positivity arguments used in the proof.

2 Preliminaries

For a real symmetric matrix M, let

$$s_+(M) = \sum_{\lambda_i(M)>0} \lambda_i(M).$$

We use the standard variational characterizations of sums of eigenvalues; see, for example, [3, 9].

We first fix the standard quotient convention used throughout; see [4, Section 2.3]. Let $\pi = (C_1, \ldots, C_t)$ be an equitable partition for a real symmetric matrix M, meaning that the row sum from a row in C_i to the columns in C_j is a constant q_{ij}. The matrix $Q = (q_{ij})$ is the *quotient matrix*; it need not itself be symmetric.

The normalized characteristic vectors of the cells give its symmetric form.

Lemma 2.1. *Let $d_i = |C_i|$, $D = \operatorname{diag}(d_1, \ldots, d_t)$, and let U have columns $\mathbf{1}_{C_i}/\sqrt{d_i}$. Then*

$$U^{\mathsf{T}} M U = D^{1/2} Q D^{-1/2}.$$

Thus Q represents the restriction of M to the cell-constant subspace and is similar to the symmetric matrix $U^\mathsf{T} MU$.

Proof. For every i, j,

$$(U^\mathsf{T} MU)_{ij} = \frac{1}{\sqrt{d_i d_j}} \sum_{u \in C_i} \sum_{v \in C_j} M_{uv} = \sqrt{\frac{d_i}{d_j}}\, q_{ij},$$

which is the (i, j)-entry of $D^{1/2} Q D^{-1/2}$. The columns of U are orthonormal and span the cell-constant subspace, proving the final assertion. □

Lemma 2.2. *If M is real symmetric and $\operatorname{tr} M = 0$, then*

$$\|M\|_* = 2s_+(M).$$

Proof. The sum of the positive eigenvalues equals the absolute value of the sum of the negative eigenvalues. Adding the two contributions gives the assertion. □

Since every distance Seidel matrix has zero diagonal, we shall use

$$E_{DS}(G) = 2s_+(D^S(G))$$

without further comment.

Ky Fan's maximum principle states that, for a real symmetric matrix M with eigenvalues $\lambda_1(M) \geq \cdots \geq \lambda_n(M)$,

$$\sum_{i=1}^{k} \lambda_i(M) = \max_{Q^\mathsf{T} Q = I_k} \operatorname{tr}(Q^\mathsf{T} MQ).$$

In particular, if C is an orthogonal compression of M, then $s_+(C) \leq s_+(M)$.

We will repeatedly use the following elementary compression lemma.

Lemma 2.3. *Let*

$$C = \begin{pmatrix} A & b \\ b^\mathsf{T} & c \end{pmatrix}$$

be a real symmetric 3×3 matrix, where A is 2×2 and $b \neq 0$. Then

$$\lambda_1(C) + \lambda_2(C) > \operatorname{tr} A.$$

Proof. The Rayleigh quotient of the third coordinate vector is c. Since $b \neq 0$, that vector is not an eigenvector, and therefore $\lambda_3(C) < c$. Hence

$$\lambda_1(C) + \lambda_2(C) = \operatorname{tr} C - \lambda_3(C) > \operatorname{tr} A.$$

□

In particular, if $\operatorname{tr} A > 0$, then $s_+(C) > \operatorname{tr} A$.

Lemma 2.4. *Let C be a real symmetric 2×2 matrix and let $t \in \mathbb{R}$. Then*

$$\det(tI - C) < 0 \quad \Longleftrightarrow \quad \lambda_2(C) < t < \lambda_1(C).$$

Proof. Since $\det(tI - C) = (t - \lambda_1(C))(t - \lambda_2(C))$, the product is negative exactly when the two factors have opposite signs. □

For a real symmetric matrix M of order N, its kth additive compound $M^{[k]}$ acts on $\bigwedge^k \mathbb{R}^N$ by

$$M^{[k]}(v_1 \wedge \cdots \wedge v_k) = \sum_{j=1}^{k} v_1 \wedge \cdots \wedge M v_j \wedge \cdots \wedge v_k.$$

Its eigenvalues are all sums $\lambda_{i_1}(M) + \cdots + \lambda_{i_k}(M)$ with $i_1 < \cdots < i_k$; see [9, Chapter 1]. Thus

$$\lambda_{\max}(M^{[k]}) = \sum_{j=1}^{k} \lambda_j(M). \tag{2.1}$$

The next elementary criterion is the only fact about additive compounds that is not used in a purely formal way.

Theorem 2.5. *Let M be a real symmetric matrix, let $1 \le k \le \dim M$, and let $L, \delta \in \mathbb{R}$. Suppose that a unit vector $w_0 \in \bigwedge^k \mathbb{R}^{\dim M}$ satisfies*

$$M^{[k]} w_0 = (L - \delta) w_0 + Y, \qquad Y \perp w_0. \tag{2.2}$$

If

$$\Xi := \|Y\|^4 + \delta\big(\langle Y, M^{[k]} Y\rangle - L\|Y\|^2\big) > 0, \tag{2.3}$$

then

$$\sum_{j=1}^{k} \lambda_j(M) > L.$$

Proof. Condition (2.3) implies $Y \neq 0$. Compress $M^{[k]}$ to the orthonormal basis $\{w_0, Y/\|Y\|\}$. By symmetry and (2.2), the compression is

$$C = \begin{pmatrix} L - \delta & \|Y\| \\ \|Y\| & \dfrac{\langle Y, M^{[k]} Y\rangle}{\|Y\|^2} \end{pmatrix}.$$

Indeed, $\langle w_0, M^{[k]} Y\rangle = \langle M^{[k]} w_0, Y\rangle = \|Y\|^2$. At the test point L,

$$\begin{aligned} \det(LI - C) &= \frac{\delta\{L\|Y\|^2 - \langle Y, M^{[k]} Y\rangle\} - \|Y\|^4}{\|Y\|^2} \\ &= -\frac{\Xi}{\|Y\|^2} < 0. \end{aligned}$$

By Lemma 2.4, $\lambda_{\max}(C) > L$. Rayleigh–Ritz and (2.1) now give

$$\sum_{j=1}^{k} \lambda_j(M) = \lambda_{\max}(M^{[k]}) \ge \lambda_{\max}(C) > L.$$

Finally, the sum of the k largest eigenvalues never exceeds the sum of all positive eigenvalues: if fewer than k eigenvalues are positive, the extra terms are nonpositive, while if at least k are positive, $s_+(M)$ contains all k terms and possibly more. Hence $s_+(M) > L$. $\square$

We also record the bilinear form used in the exterior-power calculation.

Lemma 2.6. *Let M be a real symmetric operator on a Euclidean space, and let $u_1, \dots, u_k, v_1, \dots, v_k$ be arbitrary vectors. Put $G = (\langle u_i, v_j \rangle)_{i,j=1}^k$. Then*

$$\langle u_1 \wedge \cdots \wedge u_k, v_1 \wedge \cdots \wedge v_k \rangle = \det G. \tag{2.4}$$

Moreover,

$$\Big\langle u_1 \wedge \cdots \wedge u_k, M^{[k]}(v_1 \wedge \cdots \wedge v_k) \Big\rangle = \sum_{j=1}^k \det G^{(j)}, \tag{2.5}$$

where $G^{(j)}$ is obtained from G by replacing its jth column by $(\langle u_i, Mv_j \rangle)_{i=1}^k$.

Proof. Formula (2.4) is the defining Gram-determinant formula for the Euclidean inner product on $\bigwedge^k \mathbb{R}^N$. By the definition of the additive compound,

$$M^{[k]}(v_1 \wedge \cdots \wedge v_k) = \sum_{j=1}^k v_1 \wedge \cdots \wedge Mv_j \wedge \cdots \wedge v_k.$$

Taking the inner product with $u_1 \wedge \cdots \wedge u_k$, applying (2.4) to each summand, and using linearity gives (2.5). □

3 The multipartite orthogonal decomposition

Haritha and Chithra obtained equivalent spectral information for complete multipartite graphs [8, Theorems 3.7 and 3.8]. We record the following orthogonal form because the edge-deletion argument requires explicit invariant subspaces and normalized coordinates.

The following remark separates the known spectral facts from the notation needed below.

Remark 3.1. The facts that each non-singleton part contributes a 3-eigenspace, that the singleton contrasts contribute a 1-eigenspace, and that the remaining spectrum is governed by the multipartite quotient matrix are known from [8]. What is organized here for the perturbation argument is the orthogonal normalized compression, the rank-one form (3.1), the endpoint contrast vectors, and the common invariant subspace in Lemma 3.3. No claim of priority is made for the fixed eigenvalues themselves.

Let

$$G = K_{n_1,\dots,n_q}, \qquad N = \sum_{i=1}^q n_i.$$

For each part V_i with $n_i \geq 2$, put

$$W_i = \left\{ x \in \mathbb{R}^{V(G)} : \operatorname{supp} x \subseteq V_i,\ \sum_{v \in V_i} x_v = 0 \right\}.$$

Every vector in W_i is a distance Seidel eigenvector with eigenvalue 3.

Suppose that exactly s parts are singletons. Write the sizes of the remaining parts as

$$m_1, \dots, m_\ell, \qquad m_i \geq 2.$$

The singleton contrast space

$$W_S = \left\{ x \in \mathbb{R}^{V(G)} : \operatorname{supp} x \subseteq S,\ \sum_{v \in S} x_v = 0 \right\}$$

forms an eigenspace with eigenvalue 1 and dimension $s-1$. Let

$$f_0 = \frac{\mathbf{1}_S}{\sqrt{s}} \quad (s>0), \qquad f_i = \frac{\mathbf{1}_{V_i}}{\sqrt{m_i}} \quad (1 \le i \le \ell).$$

After the fixed 3- and 1-eigenspaces are removed, the remaining symmetric matrix is

$$R = \operatorname{diag}(1, 3-2m_1, \ldots, 3-2m_\ell) - \xi\xi^{\mathsf{T}}, \qquad \xi = (\sqrt{s}, \sqrt{m_1}, \ldots, \sqrt{m_\ell})^{\mathsf{T}}, \tag{3.1}$$

where the first row and column are omitted when $s=0$.

Proposition 3.2. *The matrix R has at most one positive eigenvalue; any such eigenvalue lies in $(0,1)$.*

Proof. The diagonal matrix in (3.1) has at most one positive diagonal entry, and $\xi\xi^{\mathsf{T}}$ is positive semidefinite. Hence R has at most one positive eigenvalue. If $s=0$, every diagonal entry is negative, so no positive eigenvalue occurs. Suppose that $s>0$, and set $D_0 = \operatorname{diag}(1, 3-2m_1, \ldots, 3-2m_\ell)$. Since $R \preceq D_0$, a positive eigenvalue satisfies $\alpha \le 1$. Equality would force a corresponding unit eigenvector to lie in the 1-eigenspace of D_0 and simultaneously satisfy $\xi^{\mathsf{T}} z = 0$. That eigenspace is spanned by the singleton aggregate coordinate, whose inner product with ξ is $\sqrt{s} \neq 0$; hence equality is impossible. $\square$

When R has a positive eigenvalue α, choose a unit eigenvector z whose sign makes $\kappa = \xi^{\mathsf{T}} z > 0$. The eigenvalue equation gives

$$z_0 = \frac{\kappa\sqrt{s}}{1-\alpha}, \qquad z_i = -\frac{\kappa\sqrt{m_i}}{\alpha + 2m_i - 3}. \tag{3.2}$$

Taking its inner product with ξ and using $\|z\| = 1$ yields

$$\frac{s}{1-\alpha} = 1 + \sum_{i=1}^{\ell} \frac{m_i}{\alpha + 2m_i - 3}, \tag{3.3}$$

$$\frac{1}{\kappa^2} = \frac{s}{(1-\alpha)^2} + \sum_{i=1}^{\ell} \frac{m_i}{(\alpha + 2m_i - 3)^2}. \tag{3.4}$$

For $q \ge 3$, deleting an edge $e = uv$ changes only the distance between u and v, from 1 to 2. Consequently,

$$D^S(G-e) = D^S(G) - 2(e_u e_v^{\mathsf{T}} + e_v e_u^{\mathsf{T}}). \tag{3.5}$$

We next identify the part of the spectrum unaffected by deleting the edge. Assume $q \ge 3$ and that at most one endpoint of $e = uv$ is a singleton part. For an endpoint u in a non-singleton part A of size a, define

$$x_A = \sqrt{\frac{a-1}{a}}\, e_u - \frac{1}{\sqrt{a(a-1)}} \mathbf{1}_{A\setminus\{u\}}, \qquad W_A^0 = \{x \in W_A : x_u = 0\}.$$

If u is a singleton and $s \ge 2$, define

$$y_u = \sqrt{\frac{s-1}{s}}\, e_u - \frac{1}{\sqrt{s(s-1)}} \mathbf{1}_{S\setminus\{u\}}, \qquad W_S^0(u) = \{x \in W_S : x_u = 0\}.$$

Let $\mathcal{U}$ be the quotient space spanned by the normalized part indicators, and let $\mathcal{H}_e$ be the sum of $\mathcal{U}$ and the one distinguished vector x_A or y_u associated with each endpoint (with no extra vector when a unique singleton endpoint has $s=1$).

Lemma 3.3. *$\mathcal{H}_e$ and $\mathcal{H}_e^\perp$ are invariant under both $D^S(G)$ and $D^S(G-e)$, and the two matrices agree on $\mathcal{H}_e^\perp$. Their common restriction there is a direct sum of $3I$ and I.*

Proof. For a non-singleton endpoint,

$$e_u = \sqrt{\frac{a-1}{a}}\,x_A + \frac{1}{\sqrt{a}}f_A, \qquad W_A = \operatorname{span}\{x_A\} \oplus W_A^0.$$

For a singleton endpoint with $s \geq 2$,

$$e_u = \sqrt{\frac{s-1}{s}}\,y_u + \frac{1}{\sqrt{s}}f_0, \qquad W_S = \operatorname{span}\{y_u\} \oplus W_S^0(u).$$

Thus $e_u, e_v \in \mathcal{H}_e$. The pre-deletion matrix preserves every contrast space and the quotient space, while the perturbation in (3.5) has range contained in $\operatorname{span}\{e_u, e_v\}$ and annihilates $\mathcal{H}_e^\perp$. The last assertion follows from the fixed eigenvalues of the contrast spaces. □

4 The complete bipartite case

When $q = 2$, the connectedness of $K_{a,b} - e$ forces $a, b > 1$. We first settle this case, thereby answering the original question of Haritha and Chithra. Here the deleted edge has distance three between its endpoints, so its perturbation differs from the rank-two perturbation used when $q \geq 3$.

Proposition 4.1. *For all $a, b > 1$ and every edge e of $K_{a,b}$,*

$$E_{DS}(K_{a,b} - e) > E_{DS}(K_{a,b}).$$

Proof. Haritha and Chithra proved $E_{DS}(K_{a,b}) = 6(a + b - 2)$ and showed that the four nonfixed eigenvalues of $D^S(K_{a,b} - e)$ are the roots of the following polynomial [8, Proposition 5.1 and Theorem 5.3]:

$$\begin{aligned} F_{a,b}(x) = \; & -x^4 + (12 - 3a - 3b)x^3 + (-30 + 27a + 27b - 8ab)x^2 \\ & + (-132 - 33a - 33b + 48ab)x + 551 - 191a - 191b + 56ab. \end{aligned}$$

Let P be the sum of the positive roots of $F_{a,b}$. The remaining $a + b - 4$ eigenvalues are all equal to 3, so Lemma 2.2 gives

$$E_{DS}(K_{a,b} - e) = 6(a + b - 4) + 2P.$$

It remains to prove $P > 6$.

Substitution at the two test points gives

$$F_{a,b}(6) = 56ab - 65a - 65b - 25, \qquad F_{a,b}(7) = -128(a + b + 1).$$

By symmetry assume $2 \leq a \leq b$. If $a = 2$ and $b \geq 4$, then $F_{2,b}(6) = 47b - 155 > 0$; if $a, b \geq 3$, then $F_{a,b}(6) \geq F_{3,3}(6) = 89 > 0$. Hence, except for $(a, b) = (2, 2), (2, 3)$, the intermediate value theorem gives a positive root in $(6, 7)$, and therefore $P > 6$.

For $(a, b) = (2, 2)$, one has

$$F_{2,2}(1) = -16 < 0 < F_{2,2}(2) = 35, \qquad F_{2,2}(5) = 176 > 0 > F_{2,2}(6) = -61.$$

For $(a, b) = (2, 3)$,

$$F_{2,3}(1) = -24 < 0 < F_{2,3}(2) = 102, \qquad F_{2,3}(5) = 312 > 0 > F_{2,3}(6) = -14.$$

Thus in either exceptional case there is one positive root in $(1, 2)$ and another in $(5, 6)$; their sum is strictly greater than 6. This completes the proof. □

5 Singleton endpoints and a negative-semidefinite reduced quotient

We now assume $q \geq 3$ and use (3.5).

5.1 Both endpoints are singleton parts

Proposition 5.1. *If the endpoints of e are singleton parts, then $E_{DS}(G-e) > E_{DS}(G)$.*

Proof. Let u, v be the endpoints and put

$$w = \frac{e_u - e_v}{\sqrt{2}}, \qquad h = \frac{e_u + e_v}{\sqrt{2}}.$$

The transposition of u, v is an automorphism of G. Hence, in the decomposition $\mathbb{R}^{V(G)} = \operatorname{span}\{w\} \oplus w^{\perp}$,

$$D^S(G) = 1 \oplus B, \qquad D^S(G-e) = 3 \oplus (B - 2hh^{\mathsf{T}}).$$

Let P_+ be the positive spectral projection of B. By the Ky Fan variational principle,

$$s_+(B - 2hh^{\mathsf{T}}) \geq s_+(B) - 2\|P_+h\|^2.$$

The matrix $-D^S(G)$ is irreducible and nonnegative. Its Perron vector is strictly positive and invariant under the transposition of u, v; hence it belongs to $w^{\perp}$ and has nonzero inner product with h. Thus h has a nonzero projection on the negative spectral subspace of B, so $\|P_+h\| < 1$. Therefore

$$s_+(D^S(G-e)) - s_+(D^S(G)) \geq 2(1 - \|P_+h\|^2) > 0.$$

The result follows from Lemma 2.2. □

5.2 Exactly one endpoint is a singleton

Proposition 5.2. *If one endpoint of e is a singleton part and the other belongs to a part of size at least 2, then $E_{DS}(G-e) > E_{DS}(G)$.*

Proof. Let u be the singleton endpoint and let v belong to a part B of size $m \geq 2$. Define

$$x = \sqrt{\frac{m-1}{m}}e_v - \frac{1}{\sqrt{m(m-1)}}\mathbf{1}_{B\setminus\{v\}},$$

so that $D^S(G)x = 3x$.

Suppose first that the reduced matrix R has a positive eigenvalue α with unit eigenvector z. By (3.2), the u-coordinate and the B-coordinate of z have opposite signs, so

$$z^{\mathsf{T}}\big(D^S(G-e) - D^S(G)\big)z = -4z_uz_v > 0.$$

If u is the only singleton part, the trace of the compression to $\operatorname{span}\{x, z\}$ is strictly larger than $3+\alpha$. If there are at least two singleton parts, choose the unit singleton contrast vector y distinguished at u; then $D^S(G)y = y$, and the trace of the compression to $\operatorname{span}\{x, y, z\}$ is strictly larger than $4+\alpha$. These traces are positive, so the positive eigenvalue sum of each compression is at least its trace. By Lemma 3.3, every positive contribution outside the affected space is identical before and after deletion. The original affected contribution is respectively $3+\alpha$ or $4+\alpha$, and hence the total positive eigenvalue sum strictly increases.

It remains to consider $R \preceq 0$. Then there are at least two singleton parts. Indeed, if there were exactly one singleton part, the principal 2×2 submatrix of R on its aggregate coordinate and any non-singleton part of size m would have the form

$$\begin{pmatrix} 0 & -\sqrt{m} \\ -\sqrt{m} & 3-3m \end{pmatrix},$$

whose determinant is $-m < 0$; that principal submatrix, and hence R, would have a positive eigenvalue. Let y be as above and choose a unit eigenvector z for the smallest eigenvalue of R. Since a sufficiently large scalar multiple of the identity minus R is irreducible and entrywise positive, z may be chosen with all quotient coordinates nonzero. Moreover, $\langle x, (D^S(G-e) - D^S(G))z \rangle = -2\sqrt{(m-1)/m}\, z_u \neq 0$. In the compression to $\operatorname{span}\{x, y, z\}$ the leading 2×2 block has trace 4, and the coupling to z is nonzero. By Lemma 2.3, the positive eigenvalue sum of this compression is strictly larger than 4. Since $R \preceq 0$, the original positive contribution on the affected space is exactly 4, and Lemma 3.3 shows that all omitted fixed 3- and 1-eigenvalues cancel. The result follows from Lemma 2.2. □

5.3 Non-singleton endpoints and a negative-semidefinite reduced matrix

Proposition 5.3. *Suppose the endpoints of e lie in two parts of sizes at least 2. If the reduced matrix R is negative semidefinite, then $E_{DS}(G-e) > E_{DS}(G)$.*

Proof. Let x_A, x_B be the endpoint contrast vectors in the two parts. They are orthonormal 3-eigenvectors of $D^S(G)$. Choose a unit eigenvector z for the smallest eigenvalue of R, with all quotient coordinates nonzero as in the proof of Proposition 5.2. In the compression of $D^S(G-e)$ to $\operatorname{span}\{x_A, x_B, z\}$, the leading 2×2 block has diagonal entries $3, 3$, hence trace 6. The coupling to z is nonzero because, for example, $\langle x_A, (D^S(G-e) - D^S(G))z \rangle = -2\sqrt{(a-1)/a}\, z_v \neq 0$. By Lemma 2.3, its positive eigenvalue sum is strictly larger than 6. The original positive contribution on the affected space was exactly 6 because $R \preceq 0$. By Lemma 3.3, all remaining fixed positive eigenvalues are common to the two matrices, so Lemma 2.2 completes the proof. □

6 Only two non-singleton parts

We next treat graphs

$$K_{a,b,\underbrace{1,\ldots,1}_{s\text{ times}}}, \qquad a, b \geq 2,\ s \geq 1,$$

where the deleted edge joins the two non-singleton parts.

Put

$$p = a-1, \qquad r = b-1, \qquad S = p+r, \qquad T = pr.$$

We first derive the two reduced characteristic polynomials. Before edge deletion, use the cells consisting of the s singleton vertices, the a-part, and the b-part. After deletion, write the endpoint parts as $\{u\} \cup A'$ and $\{v\} \cup B'$, where $|A'| = p$ and $|B'| = r$, and use the five cells $\{u\}, A', \{v\}, B', S$. The corresponding quotient matrices of the distance Seidel matrix are

$$Q_0 = \begin{pmatrix} 1-s & -(p+1) & -(r+1) \\ -s & -3p & -(r+1) \\ -s & -(p+1) & -3r \end{pmatrix} \tag{6.1}$$

and

$$Q_e = \begin{pmatrix} 0 & -3p & -3 & -r & -s \\ -3 & -3(p-1) & -1 & -r & -s \\ -3 & -p & 0 & -3r & -s \\ -1 & -p & -3 & -3(r-1) & -s \\ -1 & -p & -1 & -r & -(s-1) \end{pmatrix}.$$

By Lemma 2.1, if D is the diagonal matrix of cell sizes, then $D^{1/2}Q_0D^{-1/2}$ and $D^{1/2}Q_eD^{-1/2}$ are the corresponding symmetric orthogonal compressions. Hence the two quotient matrices have the same spectra as the reduced restrictions.

We now perform the determinant expansions. From (6.1),

$$\begin{aligned}\det(xI - Q_0) = (x+s-1)\big((x+3p)(x+3r) - (p+1)(r+1)\big)\\ - s(p+1)(x+2r-1) + s(r+1)(1-2p-x).\end{aligned}$$

Collecting $S = p + r$ and $T = pr$ gives

$$\begin{aligned}\Phi_{S,T,s}(x) = x^3 + (3S+s-1)x^2 + (8T+2sS-4S-2s-1)x\\ + 4sT - 8T - 2sS + S + s + 1.\end{aligned} \tag{6.2}$$

To expand the second determinant, start from $xI - Q_e$ and replace its second row by the second row minus the first, and its fourth row by the fourth row minus the third. This gives

$$\begin{pmatrix} x & 3p & 3 & r & s \\ 3-x & x-3 & -2 & 0 & 0 \\ 3 & p & x & 3r & s \\ -2 & 0 & 3-x & x-3 & 0 \\ 1 & p & 1 & r & s+x-1 \end{pmatrix}.$$

Expanding its fifth column, and then the two sparse rows, yields

$$\det(xI - Q_e) = (x-1)H_0(x) + sH_1(x), \tag{6.3}$$

where

$$\begin{aligned} H_0(x) &= x^4 - 6x^3 + 54x - 81 + 3(p+r)x^3 - 19(p+r)x^2\\ &\quad + 21(p+r)x + 27(p+r) + 8prx^2 - 48prx + 40pr,\\ H_1(x) &= x^4 - 8x^3 + 18x^2 - 27 + 2(p+r)x^3 - 14(p+r)x^2\\ &\quad + 22(p+r)x + 6(p+r) + 4prx^2 - 24prx + 20pr.\end{aligned}$$

Substitution of $S = p + r$ and $T = pr$ into (6.3), followed only by collection of equal powers of x, gives

$$\begin{aligned}\Psi_{S,T,s}(x) = x^5 &+ (3S+s-7)x^4 + (8T+2sS-22S-8s+6)x^3\\ &+ (4sT - 56T - 14sS + 40S + 18s + 54)x^2\\ &+ (-24sT + 88T + 22sS + 6S - 135)x\\ &+ 20sT - 40T + 6sS - 27S - 27s + 81.\end{aligned} \tag{6.4}$$

Thus both reduced polynomials are obtained by explicit hand determinant expansions.

Proposition 6.1. *For every $a, b \geq 2$ and $s \geq 1$, deleting an edge between the a- and b-parts strictly increases the distance Seidel energy.*

Proof. The reduced matrix associated with (6.2) has at most one positive eigenvalue. If it has none, the result follows from Proposition 5.3. Let $\alpha > 0$ be its unique positive eigenvalue and let P be the sum of the positive roots of (6.4). Comparing the fixed 3- and 1-eigenvalues before and after deletion gives

$$E_{DS}(G - e) > E_{DS}(G) \quad \Longleftrightarrow \quad P > \alpha + 6.$$

The required root-sum inequality is proved in Appendix A by an explicit pair-sum polynomial argument. □

7 Equal endpoint parts in an arbitrary multipartite graph

Proposition 7.1. *Suppose the endpoints of e belong to two parts of the same size $m \geq 2$. Then $E_{DS}(G - e) > E_{DS}(G)$, with no restriction on the remaining parts.*

Proof. Put $p = m - 1$. Let x_A, x_B be the endpoint contrast vectors and f_A, f_B the normalized part indicators. By Lemma 3.3, all fixed 3- and 1-eigenspaces outside the affected space are common to the pre- and post-deletion matrices; they will therefore be omitted from the comparison. Set

$$x_\pm = \frac{x_A \pm x_B}{\sqrt{2}}, \qquad f_\pm = \frac{f_A \pm f_B}{\sqrt{2}}.$$

The simultaneous interchange of the two parts and the two endpoints is an automorphism of both G and $G - e$, so the symmetric and antisymmetric subspaces are invariant.

On $\operatorname{span}\{x_-, f_-\}$, the post-deletion matrix is

$$M_- = \begin{pmatrix} 3 + \dfrac{2p}{p+1} & \dfrac{2\sqrt{p}}{p+1} \\ \dfrac{2\sqrt{p}}{p+1} & 1 - 2p + \dfrac{2}{p+1} \end{pmatrix}.$$

It has exactly one positive eigenvalue

$$\lambda_p = 3 - p + \sqrt{p(p+4)}.$$

Write

$$g_p = \lambda_p - 3, \qquad h_p = g_p - \frac{2p}{p+1}, \qquad u_p = \frac{p+3}{p+1}.$$

Then $h_p > 0$. Indeed,

$$\sqrt{p(p+4)} > p + \frac{2p}{p+1},$$

and squaring reduces this inequality to $4p/(p+1)^2 > 0$.

If the reduced matrix has no positive eigenvalue, the Rayleigh quotient of x_+ after deletion is u_p, and the total gain is at least $g_p + u_p - 3 = h_p > 0$. Suppose now that the positive eigenvalue of the reduced matrix is $\alpha \in (0, 1)$, with unit eigenvector z. Put

$$\tau = \frac{|\langle f_+, z\rangle|^2}{m}.$$

The compression of the symmetric post-deletion block to $\operatorname{span}\{x_+, z\}$ is

$$B = \begin{pmatrix} u_p & -2\sqrt{\dfrac{p}{p+1}}\sqrt{\tau} \\ -2\sqrt{\dfrac{p}{p+1}}\sqrt{\tau} & \alpha - 2\tau \end{pmatrix}.$$

It suffices to show

$$s_+(B) > 3 + \alpha - g_p. \tag{7.1}$$

The positive eigenspace of the reduced matrix is one-dimensional and is invariant under interchanging the two equal parts; since the antisymmetric quotient vector has eigenvalue $3 - 2m < 0$, the vector z is symmetric on these parts. Put

$$\omega = \kappa^2, \qquad L = \alpha + 2p - 1.$$

Then $\langle f_A, z\rangle = \langle f_B, z\rangle = -\kappa\sqrt{m}/L$ and $\tau = 2\omega/L^2$. A positive eigenvalue of the reduced matrix forces at least one singleton part. Combining (3.3) and (3.4), and retaining only the two equal parts, gives

$$\frac{1}{\omega} \geq \frac{1 + 2m/L}{1-\alpha} + \frac{2m}{L^2} = \frac{L^2 + 4p(p+1)}{(1-\alpha)L^2}.$$

Moreover,

$$L^2 + 4p(p+1) = L(\alpha + 4p + 1) + 2(p+1)(1-\alpha) > L(\alpha + 4p + 1).$$

Consequently,

$$\tau < U_p(\alpha) := \frac{2(1-\alpha)}{(\alpha + 2p - 1)(\alpha + 4p + 1)}.$$

Writing $A = \alpha + 2p - 1$ and $B = \alpha + 4p + 1$, we have the explicit derivative

$$U_p'(\alpha) = -\frac{2\{AB + (1-\alpha)(A+B)\}}{A^2B^2} < 0.$$

Set $\alpha_0 = 3h_p/u_p$. We claim that

$$\alpha \geq \alpha_0 \quad \Longrightarrow \quad \tau < h_p/2. \tag{7.2}$$

If $\alpha_0 \geq 1$, this implication is vacuous because $\alpha < 1$. Accordingly, whenever the implication has content we may use the monotonicity of U_p throughout $[\alpha_0, 1)$. For $p \geq 2$, put $g = g_p$ and

$$\eta = (p+1)g - 2p, \qquad h_p = \frac{\eta}{p+1}, \qquad \alpha_0 = \frac{3\eta}{p+3}.$$

The defining radical gives the quadratic relation

$$g^2 + 2pg - 4p = 0. \tag{7.3}$$

Introduce the three linear expressions

$$X = 3(p+1)g + 2p^2 - p - 3,$$
$$Y = 3(p+1)g + 4p^2 + 7p + 3,$$
$$Z = 7p + 3 - 3(p+1)g.$$

Direct substitution of $\alpha_0 = 3\eta/(p+3)$ gives

$$\alpha_0 + 2p - 1 = \frac{X}{p+3}, \qquad \alpha_0 + 4p + 1 = \frac{Y}{p+3}, \qquad 1 - \alpha_0 = \frac{Z}{p+3}.$$

Consequently,

$$\frac{h_p}{2} - U_p(\alpha_0) = \frac{\eta XY - 4(p+1)(p+3)Z}{2(p+1)XY}. \tag{7.4}$$

Only the quadratic relation for the radical is needed: replace every occurrence of g^2 by $4p - 2pg$ using (7.3). Multiplying out the two sides and performing this single reduction gives

$$XY = (p+1)^2(8p^2 + 30p - 9), \tag{7.5}$$

$$\eta XY - 4(p+1)(p+3)Z = (p+1)(\mathsf{A}_p g - \mathsf{B}_p), \tag{7.6}$$

where

$$\mathsf{A}_p = 8p^4 + 46p^3 + 71p^2 + 60p + 27,$$
$$\mathsf{B}_p = 16p^4 + 76p^3 + 70p^2 + 78p + 36.$$

For example, (7.5) follows from

$$XY = 9(p+1)^2 g^2 + 6(p+1)(3p^2 + 3p)g + (2p^2 - p - 3)(4p^2 + 7p + 3),$$

after replacing g^2 by $4p - 2pg$; the coefficient of g cancels. Once (7.5) is known, the second reduction contains no hidden step: with $D = 8p^2 + 30p - 9$,

$$\begin{aligned}
&\eta XY - 4(p+1)(p+3)Z \\
&= (p+1)\Big(\big[(p+1)^2 D + 12(p+1)(p+3)\big]g \\
&\qquad\qquad - \big[2p(p+1)D + 4(p+3)(7p+3)\big]\Big),
\end{aligned}$$

and the two bracketed coefficients expand respectively to A_p and B_p. Substitution of (7.5)–(7.6) into (7.4) yields

$$\frac{h_p}{2} - U_p(\alpha_0) = \frac{\mathsf{A}_p g_p - \mathsf{B}_p}{2(p+1)^2(8p^2 + 30p - 9)}.$$

All denominator factors are positive. Since $g_p = \sqrt{p(p+4)} - p$, the numerator is positive precisely when

$$\mathsf{A}_p \sqrt{p(p+4)} > \mathsf{A}_p\, p + \mathsf{B}_p.$$

Both sides are positive. Squaring is therefore equivalent, and direct expansion followed by the shift $p = t + 2$ gives

$$\begin{aligned}
&\mathsf{A}_p^2 p(p+4) - (\mathsf{A}_p\, p + \mathsf{B}_p)^2 \\
&\quad = 4\big(48t^7 + 1060t^6 + 9478t^5 + 44424t^4 + 117765t^3 + 174836t^2 + 130347t + 33962\big) > 0.
\end{aligned}$$

Every coefficient in the last polynomial is positive and $t = p - 2 \geq 0$. Thus $U_p(\alpha_0) < h_p/2$, and the monotonicity of U_p proves (7.2) for $p \geq 2$.

It remains to justify (7.2) for $p = 1$. Put

$$r_0 = \sqrt{5} - 2 = h_1, \qquad \alpha_0 = \frac{3}{2} r_0, \qquad u = |\langle f_+, z\rangle|^2 = 2\tau.$$

Let σ denote the number of singleton parts. The secular equation implies

$$\sigma \geq h(\alpha) := \frac{(1-\alpha)(\alpha+5)}{\alpha+1},$$

whereas normalization gives

$$u \leq U_\sigma(\alpha) := \frac{4(1-\alpha)^2}{\sigma(1+\alpha)^2 + 4(1-\alpha)^2}.$$

Set

$$\alpha_2 = -3 + 2\sqrt{3}, \qquad \alpha_1 = \frac{-5+\sqrt{41}}{2}.$$

These thresholds satisfy

$$0 < \alpha_0 < \alpha_2 < \alpha_1 < 1.$$

Indeed, $\alpha_0 < \alpha_2$ is equivalent to $3\sqrt{5}/2 < 2\sqrt{3}$, whose squared form is $45 < 48$; $\alpha_2 < \alpha_1$ follows from $(4\sqrt{3}-1)^2 < 41$, which is equivalent to $1 < \sqrt{3}$; and $\alpha_1 < 1$ follows from $\sqrt{41} < 7$. The required monotonicity follows by differentiation. With $D_\sigma(\alpha) = \sigma(1+\alpha)^2 + 4(1-\alpha)^2$,

$$h'(\alpha) = -\frac{\alpha^2 + 2\alpha + 9}{(\alpha+1)^2} < 0,$$

$$\frac{\partial U_\sigma}{\partial \alpha} = -\frac{16\sigma(1-\alpha)(1+\alpha)}{D_\sigma(\alpha)^2} < 0, \qquad \frac{\partial U_\sigma}{\partial \sigma} = -\frac{4(1-\alpha)^2(1+\alpha)^2}{D_\sigma(\alpha)^2} < 0.$$

Also $h(\alpha_2) = 2$ and $h(\alpha_1) = 1$. Hence

$$\sigma \geq 3 \quad (\alpha_0 \leq \alpha < \alpha_2), \qquad \sigma \geq 2 \quad (\alpha_2 \leq \alpha < \alpha_1),$$

and $\sigma \geq 1$ on $[\alpha_1, 1)$. The endpoint values are

$$\begin{aligned}
U_3(\alpha_0) &= \frac{16444 - 3744\sqrt{5}}{34681} < \frac{7}{30} < r_0,\\
U_2(\alpha_2) &= \frac{1}{2} - \frac{\sqrt{3}}{6} < \frac{7}{30} < r_0,\\
U_1(\alpha_1) &= \frac{1}{2} - \frac{5\sqrt{41}}{82} < \frac{1}{5} < r_0.
\end{aligned}$$

For complete arithmetic verification, the first strict inequality is equivalent to $112320\sqrt{5} > 250553$, whose squared sides differ by $302106191 > 0$. The remaining comparisons reduce respectively to $75 > 64$, $4500 > 4489$, $25625 > 15129$, and $125 > 121$ after squaring positive quantities. Since U_σ decreases in both arguments, these bounds give $u < r_0 = h_1$, or equivalently $\tau < h_1/2$, which proves (7.2) for $p = 1$.

Now $s_+(B) \geq u_p$ and $s_+(B) \geq \operatorname{tr} B = u_p + \alpha - 2\tau$. If $\alpha < h_p$, the first bound gives (7.1). If $\tau < h_p/2$, the second gives it. In the remaining case $\alpha \geq h_p$ and $\tau \geq h_p/2$, relation (7.2) implies $\alpha < \alpha_0$. Also $\alpha_0 < g_p$: after clearing the positive denominator $p+3$, this is equivalent to $g_p < 3$. In fact,

$$g_p = \sqrt{p(p+4)} - p = \frac{4p}{\sqrt{p(p+4)} + p} < 2.$$

Hence $g_p - \alpha > 0$. For $T_0 = 3 + \alpha - g_p$, direct expansion of the 2×2 determinant gives

$$\det(T_0 I - B) = (3 - g_p)(\alpha - h_p) - 2\tau(g_p - \alpha).$$

Since $\tau \ge h_p/2$ in the remaining case,

$$\begin{aligned}\det(T_0 I - B) &\le (3-g_p)(\alpha - h_p) - h_p(g_p - \alpha)\\ &= \big(3 - g_p + h_p\big)\alpha - 3h_p = u_p\alpha - 3h_p < 0,\end{aligned}$$

where the last inequality is exactly $\alpha < \alpha_0 = 3h_p/u_p$. By Lemma 2.4, the largest eigenvalue of B is larger than T_0, proving (7.1). Together with the unchanged fixed spectral contribution from Lemma 3.3, this proves the proposition. □

8 Unequal endpoint parts

It remains to consider the following situation:

- the endpoints lie in two non-singleton parts of unequal sizes $2 \le a < b$;
- the reduced matrix has a positive eigenvalue $\alpha \in (0,1)$;
- besides the endpoint parts, there is at least one further non-singleton part.

The last condition is precisely what remains after Propositions 6.1 and 7.1.

Let A and B be the endpoint parts, let f_A, f_B be their normalized indicators in the quotient space $\mathcal{U}$, and let x_A, x_B be the corresponding endpoint contrast vectors from Lemma 3.3. Put

$$c_a = \sqrt{\frac{a-1}{a}}, \qquad c_b = \sqrt{\frac{b-1}{b}}.$$

Thus

$$e_u = c_a x_A + \frac{1}{\sqrt{a}} f_A, \qquad e_v = c_b x_B + \frac{1}{\sqrt{b}} f_B. \tag{8.1}$$

Let $z \in \mathcal{U}$ be the unit eigenvector belonging to α, with its sign chosen as in Proposition 3.2, and write

$$\omega = \kappa^2, \qquad L_a = \alpha + 2a - 3, \qquad L_b = \alpha + 2b - 3.$$

Then

$$\mu_a := \langle f_A, z\rangle = -\frac{\sqrt{a\omega}}{L_a}, \qquad \mu_b := \langle f_B, z\rangle = -\frac{\sqrt{b\omega}}{L_b}.$$

Let M denote the restriction of $D^S(G-e)$ to the affected space

$$\mathcal{H}_e = \operatorname{span}\{x_A, x_B\} \oplus \mathcal{U}.$$

With respect to this orthogonal direct sum, (3.5) and (8.1) give the exact block form

$$M = \begin{pmatrix} 3 & -2c_a c_b & -\frac{2c_a}{\sqrt{b}} f_B^\mathsf{T} \\ -2c_a c_b & 3 & -\frac{2c_b}{\sqrt{a}} f_A^\mathsf{T} \\ -\frac{2c_a}{\sqrt{b}} f_B & -\frac{2c_b}{\sqrt{a}} f_A & R - \frac{2}{\sqrt{ab}}(f_A f_B^\mathsf{T} + f_B f_A^\mathsf{T}) \end{pmatrix}. \tag{8.2}$$

Before deletion, the restriction to $\mathcal{H}_e$ is $3 \oplus 3 \oplus R$. Since R has the unique positive eigenvalue α, its positive eigenvalue sum is therefore $6 + \alpha$. By Lemma 3.3, it suffices to prove

$$s_+(M) > 6 + \alpha.$$

8.1 The third-additive-compound compression

Set

$$w_0 = x_A \wedge x_B \wedge z, \qquad L = 6 + \alpha, \qquad \delta = \frac{4\omega}{L_a L_b}.$$

The vector w_0 is a unit vector. Define

$$Y = (I - w_0 w_0^{\mathsf{T}}) M^{[3]} w_0.$$

The following expansion makes the additive-compound calculation completely explicit.

Put

$$r_A = f_A - \mu_a z, \qquad r_B = f_B - \mu_b z, \qquad \rho_a = 1 - \mu_a^2, \qquad \rho_b = 1 - \mu_b^2.$$

Although r_A and r_B need not be orthogonal, they are both orthogonal to z, and

$$\langle r_A, r_A \rangle = \rho_a, \qquad \langle r_B, r_B \rangle = \rho_b, \qquad \langle r_A, r_B \rangle = -\mu_a \mu_b, \tag{8.3}$$

$$\langle r_A, R r_A \rangle = -3(a-1) - \alpha \mu_a^2, \qquad \langle r_B, R r_B \rangle = -3(b-1) - \alpha \mu_b^2, \tag{8.4}$$

$$\langle r_A, R r_B \rangle = -\sqrt{ab} - \alpha \mu_a \mu_b.$$

The first line follows from $\langle f_A, f_B \rangle = 0$. For the other lines, the reduced matrix has

$$\langle f_A, R f_A \rangle = -3(a-1), \qquad \langle f_B, R f_B \rangle = -3(b-1), \qquad \langle f_A, R f_B \rangle = -\sqrt{ab}.$$

Using $Rz = \alpha z$, for example,

$$\begin{aligned} \langle r_A, R r_A \rangle &= \langle f_A, R f_A \rangle - 2\mu_a \langle f_A, R z \rangle + \mu_a^2 \langle z, R z \rangle \\ &= -3(a-1) - \alpha \mu_a^2, \end{aligned}$$

and the other two identities follow by the same expansion.

From (8.2) one obtains

$$M x_A = 3 x_A - 2 c_a c_b x_B - \frac{2 c_a}{\sqrt{b}} f_B, \tag{8.5}$$

$$M x_B = 3 x_B - 2 c_a c_b x_A - \frac{2 c_b}{\sqrt{a}} f_A,$$

$$M z = \left(\alpha - \frac{4\omega}{L_a L_b} \right) z - \frac{2 c_a \mu_b}{\sqrt{b}} x_A - \frac{2 c_b \mu_a}{\sqrt{a}} x_B - \frac{2}{\sqrt{ab}} (\mu_b r_A + \mu_a r_B). \tag{8.6}$$

By definition,

$$M^{[3]} w_0 = (M x_A) \wedge x_B \wedge z + x_A \wedge (M x_B) \wedge z + x_A \wedge x_B \wedge (M z).$$

Substituting (8.5)–(8.6), writing $f_A = \mu_a z + r_A$ and $f_B = \mu_b z + r_B$, and deleting every wedge with a repeated factor gives

$$M^{[3]} w_0 = (L - \delta) w_0 + Y, \qquad Y = 2 \sum_{i=1}^{4} d_i E_i, \tag{8.7}$$

where

$$E_1 = r_B \wedge x_B \wedge z, \qquad E_2 = x_A \wedge r_A \wedge z,$$

$$E_3 = x_A \wedge x_B \wedge r_A, \qquad E_4 = x_A \wedge x_B \wedge r_B,$$

and

$$d = \begin{pmatrix} -c_a/\sqrt{b} \\ -c_b/\sqrt{a} \\ -\mu_b/\sqrt{ab} \\ -\mu_a/\sqrt{ab} \end{pmatrix}.$$

Let

$$\Gamma_{ij} = \langle E_i, E_j \rangle, \qquad K_{ij} = \langle E_i, (LI - M^{[3]})E_j \rangle.$$

The Gram-determinant formula in Lemma 2.6 and (8.3) give

$$\Gamma = \begin{pmatrix} \rho_b & 0 & 0 & 0 \\ 0 & \rho_a & 0 & 0 \\ 0 & 0 & \rho_a & -\mu_a\mu_b \\ 0 & 0 & -\mu_a\mu_b & \rho_b \end{pmatrix}. \tag{8.8}$$

We next derive K entry by entry. Write

$$M = M_0 + \Delta, \qquad M_0 = 3I_{\operatorname{span}\{x_A, x_B\}} \oplus R, \qquad \Delta = -2(e_u e_v^{\mathsf{T}} + e_v e_u^{\mathsf{T}}).$$

First consider the contribution of M_0. For example,

$$\begin{aligned} \langle E_1, M_0^{[3]} E_1 \rangle &= \langle r_B, R r_B \rangle + (3+\alpha)\rho_b, \\ L\rho_b - \langle E_1, M_0^{[3]} E_1 \rangle &= 3b + (\alpha - 3)\mu_b^2, \end{aligned}$$

where (8.4) is used in the second line. Similarly,

$$\begin{aligned} L\rho_a - \langle E_3, M_0^{[3]} E_3 \rangle &= 3(a-1) + \alpha, \\ L\langle E_3, E_4 \rangle - \langle E_3, M_0^{[3]} E_4 \rangle &= \sqrt{ab}; \end{aligned}$$

the last identity follows from

$$\langle E_3, M_0^{[3]} E_4 \rangle = 6\langle r_A, r_B \rangle + \langle r_A, R r_B \rangle = -\sqrt{ab} - L\mu_a\mu_b.$$

All other off-diagonal contributions of M_0 vanish. Hence its shifted contribution is

$$K^{(0)} = \begin{pmatrix} 3b + (\alpha-3)\mu_b^2 & 0 & 0 & 0 \\ 0 & 3a + (\alpha-3)\mu_a^2 & 0 & 0 \\ 0 & 0 & 3(a-1)+\alpha & \sqrt{ab} \\ 0 & 0 & \sqrt{ab} & 3(b-1)+\alpha \end{pmatrix}. \tag{8.9}$$

For the perturbation, the endpoint inner products required by Lemma 2.6 are

	x_A	x_B	z	r_A	r_B
$\langle \cdot, e_u \rangle$	c_a	0	$\mu_a/\sqrt{a}$	$\rho_a/\sqrt{a}$	$-\mu_a\mu_b/\sqrt{a}$
$\langle \cdot, e_v \rangle$	0	c_b	$\mu_b/\sqrt{b}$	$-\mu_a\mu_b/\sqrt{b}$	$\rho_b/\sqrt{b}$

(8.10)

More explicitly, if $E = u_1 \wedge u_2 \wedge u_3$ and $F = v_1 \wedge v_2 \wedge v_3$, then

$$-\langle E, \Delta^{[3]} F \rangle = 2\sum_{j=1}^{3} \big(\langle e_v, v_j \rangle \det G_j(e_u) + \langle e_u, v_j \rangle \det G_j(e_v) \big), \tag{8.11}$$

where $G_j(e)$ is the Gram matrix $(\langle u_i, v_k\rangle)$ with its jth column replaced by $(\langle u_i, e\rangle)_{i=1}^3$. Formula (8.11) follows immediately from $\Delta v = -2(e_u\langle e_v, v\rangle + e_v\langle e_u, v\rangle)$ and Lemma 2.6.

Substituting (8.10) into (8.11) gives

$$K^{(\Delta)} = \begin{pmatrix} 0 & 2c_a c_b \mu_a \mu_b & \dfrac{2c_a\mu_a\mu_b^2}{\sqrt{b}} & -\dfrac{2c_a\mu_b\rho_b}{\sqrt{b}} \\ 2c_a c_b\mu_a\mu_b & 0 & -\dfrac{2c_b\mu_a\rho_a}{\sqrt{a}} & \dfrac{2c_b\mu_a^2\mu_b}{\sqrt{a}} \\ \dfrac{2c_a\mu_a\mu_b^2}{\sqrt{b}} & -\dfrac{2c_b\mu_a\rho_a}{\sqrt{a}} & -\dfrac{4\mu_a\mu_b\rho_a}{\sqrt{ab}} & \dfrac{4\mu_a^2\mu_b^2 - 2\mu_a^2 - 2\mu_b^2 + 2}{\sqrt{ab}} \\ -\dfrac{2c_a\mu_b\rho_b}{\sqrt{b}} & \dfrac{2c_b\mu_a^2\mu_b}{\sqrt{a}} & \dfrac{4\mu_a^2\mu_b^2 - 2\mu_a^2 - 2\mu_b^2 + 2}{\sqrt{ab}} & -\dfrac{4\mu_a\mu_b\rho_b}{\sqrt{ab}} \end{pmatrix}. \tag{8.12}$$

For clarity, we record all nonredundant determinant sums. Applying (8.11) column by column and then using $\rho_a = 1-\mu_a^2$ and $\rho_b = 1-\mu_b^2$ gives

$$\begin{aligned}
K_{11}^{(\Delta)} &= \frac{4\mu_a\mu_b(\mu_b^2-1)}{\sqrt{ab}} - \frac{4\mu_a\mu_b(\mu_b^2-1)}{\sqrt{ab}} = 0,\\
K_{12}^{(\Delta)} &= 2c_a c_b\mu_a\mu_b,\\
K_{13}^{(\Delta)} &= \frac{2c_a\mu_a\mu_b^2}{\sqrt{b}},\\
K_{14}^{(\Delta)} &= \frac{2c_a\mu_b(\mu_b^2-1)}{\sqrt{b}} = -\frac{2c_a\mu_b\rho_b}{\sqrt{b}},\\
K_{33}^{(\Delta)} &= \frac{4\mu_a\mu_b(\mu_a^2-1)}{\sqrt{ab}} = -\frac{4\mu_a\mu_b\rho_a}{\sqrt{ab}},\\
K_{34}^{(\Delta)} &= \frac{2(2\mu_a^2\mu_b^2 - \mu_a^2 - \mu_b^2 + 1)}{\sqrt{ab}}.
\end{aligned}$$

The simultaneous interchange $A \leftrightarrow B$ sends (E_1, E_2, E_3, E_4) to $(-E_2, -E_1, -E_4, -E_3)$; it therefore gives $K_{22}^{(\Delta)}, K_{23}^{(\Delta)}, K_{24}^{(\Delta)}$, and $K_{44}^{(\Delta)}$ from the six formulas above. The lower triangle follows from symmetry. Hence every entry in (8.12) has now been accounted for directly from (8.11). Adding (8.9) and (8.12), we obtain the symmetric matrix K whose upper-triangular entries are

$$K_{11} = 3b + (\alpha-3)\mu_b^2, \qquad K_{22} = 3a + (\alpha-3)\mu_a^2, \tag{8.13}$$

$$K_{12} = 2c_a c_b\mu_a\mu_b, \qquad K_{13} = \frac{2c_a\mu_a\mu_b^2}{\sqrt{b}}, \qquad K_{14} = -\frac{2c_a\mu_b\rho_b}{\sqrt{b}},$$

$$K_{23} = -\frac{2c_b\mu_a\rho_a}{\sqrt{a}}, \qquad K_{24} = \frac{2c_b\mu_a^2\mu_b}{\sqrt{a}},$$

$$K_{33} = 3(a-1) + \alpha - \frac{4\mu_a\mu_b\rho_a}{\sqrt{ab}}, \qquad K_{44} = 3(b-1) + \alpha - \frac{4\mu_a\mu_b\rho_b}{\sqrt{ab}},$$

$$K_{34} = \sqrt{ab} + \frac{4\mu_a^2\mu_b^2 - 2\mu_a^2 - 2\mu_b^2 + 2}{\sqrt{ab}}. \tag{8.14}$$

For later use, define

$$\mathcal{B} = \frac{a+b-2}{ab} - \frac{(b-2)\omega}{bL_a^2} - \frac{(a-2)\omega}{aL_b^2} - \frac{4\omega^2}{L_a^2 L_b^2}, \tag{8.15}$$

$$\begin{aligned}\mathcal{G} = 6 - \frac{3}{a} - \frac{3}{b} + \alpha\omega\left(\frac{1}{L_a^2} + \frac{1}{L_b^2}\right) + \frac{6\omega}{L_a L_b} - \frac{8\omega}{L_a L_b}\left(\frac{1}{a} + \frac{1}{b}\right)\\ + \frac{8\omega^2}{L_a L_b}\left(\frac{1}{L_a^2} + \frac{1}{L_b^2}\right) + \frac{16\omega}{abL_a L_b}\left(1 - \frac{a\omega}{L_a^2}\right)\left(1 - \frac{b\omega}{L_b^2}\right).\end{aligned} \tag{8.16}$$

and

$$\mathcal{N} := \|Y\|^4 + \delta\big(\langle Y, M^{[3]}Y\rangle - L\|Y\|^2\big). \tag{8.17}$$

The exterior-power calculation now reduces to two quadratic forms.

Lemma 8.1. *The following identities hold:*

$$\|Y\|^2 = 4d^{\mathsf{T}}\Gamma d = 4\mathcal{B}, \qquad L\|Y\|^2 - \langle Y, M^{[3]}Y\rangle = 4d^{\mathsf{T}}Kd = 4\mathcal{G}. \tag{8.18}$$

$$\mathcal{N} = 16\left(\mathcal{B}^2 - \frac{\omega\mathcal{G}}{L_a L_b}\right). \tag{8.19}$$

Proof. By (8.7), $Y = 2\sum_i d_i E_i$. Hence

$$\|Y\|^2 = 4d^{\mathsf{T}}\Gamma d, \qquad L\|Y\|^2 - \langle Y, M^{[3]}Y\rangle = 4d^{\mathsf{T}}Kd.$$

We first expand the Gram form. From (8.8),

$$\begin{aligned}d^{\mathsf{T}}\Gamma d &= \frac{c_a^2}{b}\rho_b + \frac{c_b^2}{a}\rho_a + \frac{\mu_b^2}{ab}\rho_a + \frac{\mu_a^2}{ab}\rho_b - \frac{2\mu_a^2\mu_b^2}{ab}\\ &= \frac{a+b-2}{ab} - \frac{a-2}{ab}\mu_b^2 - \frac{b-2}{ab}\mu_a^2 - \frac{4\mu_a^2\mu_b^2}{ab}.\end{aligned}$$

Using $\mu_a^2 = a\omega/L_a^2$ and $\mu_b^2 = b\omega/L_b^2$ gives exactly (8.15).

For the shifted form, substitute the entries (8.13)–(8.14) into $\sum_i d_i^2 K_{ii} + 2\sum_{i<j} d_i d_j K_{ij}$, and then use $c_a^2 = (a-1)/a$, $c_b^2 = (b-1)/b$ and $\mu_a = -\sqrt{a\omega}/L_a$, $\mu_b = -\sqrt{b\omega}/L_b$. Grouping by powers of ω gives

$$\begin{aligned}d^{\mathsf{T}}Kd = 6 - \frac{3}{a} - \frac{3}{b}\\ + \omega\left[\alpha\left(\frac{1}{L_a^2} + \frac{1}{L_b^2}\right) + \frac{6}{L_a L_b} - \frac{8}{L_a L_b}\left(\frac{1}{a} + \frac{1}{b}\right) + \frac{16}{abL_a L_b}\right]\\ + \omega^2\left[\frac{8}{L_a^3 L_b} + \frac{8}{L_a L_b^3} - \frac{16}{bL_a^3 L_b} - \frac{16}{aL_a L_b^3}\right] + \frac{16\omega^3}{L_a^3 L_b^3}.\end{aligned} \tag{8.20}$$

The last term in the first square bracket, the two negative terms in the second square bracket, and the cubic term are precisely the expansion of

$$\frac{16\omega}{abL_a L_b}\left(1 - \frac{a\omega}{L_a^2}\right)\left(1 - \frac{b\omega}{L_b^2}\right).$$

The two remaining quadratic terms in (8.20) equal $8\omega^2(L_a^{-2} + L_b^{-2})/(L_a L_b)$. Therefore $d^{\mathsf{T}}Kd = \mathcal{G}$, proving (8.18).

Finally, (8.17), $\delta = 4\omega/(L_a L_b)$, and (8.18) give

$$\begin{aligned} \mathcal{N} &= (4\mathcal{B})^2 - \frac{4\omega}{L_a L_b}(4\mathcal{G}) \\ &= 16\left(\mathcal{B}^2 - \frac{\omega\mathcal{G}}{L_a L_b}\right), \end{aligned}$$

which is (8.19). □

Equations (8.7) and (8.17) place the present situation exactly within Theorem 2.5, with $k = 3$ and $L = 6 + \alpha$. Therefore

$$\mathcal{N} > 0 \quad \Longrightarrow \quad \lambda_1(M) + \lambda_2(M) + \lambda_3(M) > 6 + \alpha \quad \Longrightarrow \quad s_+(M) > 6 + \alpha. \tag{8.21}$$

Thus the entire asymmetric case is reduced to proving $\mathcal{N} > 0$.

Because a further non-singleton part is present, its summand $n/(\alpha + 2n - 3)$ in (3.3) is strictly larger than $1/2$. Using (3.3) to eliminate the singleton term from (3.4), and then discarding all positive normalization summands except those of the two endpoint parts, gives

$$\begin{aligned} \frac{1}{\omega} &= \frac{1 + \sum_i n_i/(\alpha + 2n_i - 3)}{1 - \alpha} + \sum_i \frac{n_i}{(\alpha + 2n_i - 3)^2} \\ &> \frac{\frac{3}{2} + a/L_a + b/L_b}{1 - \alpha} + \frac{a}{L_a^2} + \frac{b}{L_b^2}. \end{aligned}$$

Taking reciprocals of these positive quantities gives

$$0 < \omega < \Omega_{a,b}(\alpha) := \left[\frac{\frac{3}{2} + \frac{a}{L_a} + \frac{b}{L_b}}{1 - \alpha} + \frac{a}{L_a^2} + \frac{b}{L_b^2}\right]^{-1}. \tag{8.22}$$

It remains only to verify the following exact inequality. Its entirely algebraic proof is given in Appendix B.

Lemma 8.2. *For integers $2 \le a < b$, $0 < \alpha < 1$, and ω arising from the final asymmetric configuration above, the quantity $\mathcal{N}$ in (8.19) is strictly positive.*

By Lemma 8.2 and (8.21), the positive eigenvalue sum on the affected space strictly exceeds $6 + \alpha$. The fixed positive contribution on $\mathcal{H}_e^\perp$ is unchanged by Lemma 3.3. Hence edge deletion strictly increases the distance Seidel energy in the final asymmetric case.

9 Completion of the proof

Proof of Theorem 1.2. If $q = 2$, apply Proposition 4.1. Assume $q \ge 3$. If an endpoint of e is a singleton part, apply Propositions 5.1 and 5.2. Thus both endpoint parts have size at least 2.

Let R be the reduced matrix. If $R \preceq 0$, apply Proposition 5.3. Otherwise R has a unique positive eigenvalue. If there are only two non-singleton parts, apply Proposition 6.1. If the endpoint parts have equal size, apply Proposition 7.1. The only remaining case has unequal endpoint parts and an additional non-singleton part, and is settled by Lemma 8.2 and the argument of Section 8. All possibilities are exhausted. □

Corollary 9.1. *The question of Haritha and Chithra for complete bipartite graphs has an affirmative answer: for all $a, b > 1$ and every edge e of $K_{a,b}$,*

$$E_{DS}(K_{a,b} - e) > E_{DS}(K_{a,b}).$$

10 A counterexample for matching deletion

Proof of Proposition 1.3. Let the parts of $K_{1,3,3}$ be $\{w\}$, A, and B. Before deletion, the two contrast spaces supported on A and B give the eigenvalue 3 with multiplicity 4. On the normalized part-indicator space, the vector $(f_A - f_B)/\sqrt{2}$ has eigenvalue -3, while the restriction to $\operatorname{span}\{e_w, (f_A + f_B)/\sqrt{2}\}$ is

$$\begin{pmatrix} 0 & -\sqrt{6} \\ -\sqrt{6} & -9 \end{pmatrix}.$$

Its eigenvalues are $(-9 \pm \sqrt{105})/2$. Hence

$$E_{DS}(K_{1,3,3}) = 15 + \sqrt{105}. \tag{10.1}$$

Now label $A = \{a_1, a_2, a_3\}$ and $B = \{b_1, b_2, b_3\}$, and delete $M = \{a_i b_i : 1 \leq i \leq 3\}$. On the subspace where the coordinates on each of A and B sum to zero, the distance Seidel matrix has the block form

$$\begin{pmatrix} 3I & -2I \\ -2I & 3I \end{pmatrix}.$$

It therefore contributes the eigenvalues 1 and 5, each with multiplicity 2. On the normalized part-indicator space, $(f_A - f_B)/\sqrt{2}$ has eigenvalue -1, and the remaining two-dimensional restriction is

$$\begin{pmatrix} 0 & -\sqrt{6} \\ -\sqrt{6} & -11 \end{pmatrix},$$

whose eigenvalues are $(-11 \pm \sqrt{145})/2$. Consequently,

$$E_{DS}(K_{1,3,3} - M) = 13 + \sqrt{145}. \tag{10.2}$$

Finally, $\sqrt{145} < \sqrt{105} + 2$ because $\sqrt{105} > 9$; comparison of (10.1) and (10.2) proves the claim. $\square$

11 Concluding remarks

The theorem gives a class-wide strict monotonicity result for distance Seidel energy on a natural dense graph family. It also illustrates why energy questions are substantially subtler than spectral-radius questions: edge deletion produces a rank-two perturbation of mixed sign, so neither entrywise comparison nor Loewner order controls the trace norm. The orthogonal decomposition and additive-compound method may be useful for the following problems.

(i) Characterize the matchings whose deletion increases the distance Seidel energy of a complete multipartite graph.

(ii) Determine the extremal connected graphs of fixed order and size for E_{DS}.

(iii) Quantify the minimum possible increment $E_{DS}(G - e) - E_{DS}(G)$ among complete multipartite graphs of fixed order.

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Statements and declarations

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Declaration of generative AI and AI-assisted technologies. We used ChatGPT to brainstorm the structure, anticipate potential objections, and refine the text. We are fully responsible for all claims, citations, calculations, and the final wording.

A Root-sum inequalities for two non-singleton parts

We prove the root-sum assertion used in Proposition 6.1. We first derive the pair-sum polynomial from Newton identities and then carry out the required polynomial divisions by an explicit three-term recurrence. Every step is a finite identity in $\mathbb{Z}[S, T, \alpha]$; the complete coefficient calculations are supplied in Supplementary File S1.

Recall that

$$S = p + r, \qquad T = pr,$$

where $p, r \geq 1$. Hence

$$S \geq 2, \qquad S - 1 \leq T \leq \frac{S^2}{4}. \tag{A.1}$$

The lower bound follows from $(p-1)(r-1) \geq 0$, and the upper bound from $4pr \leq (p+r)^2$.

The reduced cubic (6.2) has at most one positive root, counted with multiplicity, and

$$\Phi_{S,T,s}(0) = 4(s-2)T - (2s-1)S + s + 1.$$

If this value is negative, monicity gives a positive root. If it is positive, a positive root would either have even multiplicity or force a second positive zero by a sign change before the polynomial returns to $+\infty$; both alternatives contradict the one-positive-eigenvalue property. The equality cases must be checked separately. A direct factorization gives

$$\begin{aligned}
\Phi_{4,4,3}(x) &= x(x+3)(x+11), && (s,p,r) = (3,2,2),\\
\Phi_{3,2,4}(x) &= x(x^2+12x+19), && (s,p,r) = (4,1,2) \text{ or } (4,2,1),\\
\Phi_{2,1,5}(x) &= x(x+1)(x+9), && (s,p,r) = (5,1,1).
\end{aligned}$$

Here the subscripts of Φ are (S, T, s). None of these boundary polynomials has a positive root. Consequently, a positive root of the reduced cubic occurs precisely in the following cases:

s	parameters for which Φ has a positive root
$1, 2$	all $p, r \geq 1$,
3	$\min\{p, r\} = 1$,
4	$p = r = 1$,
$s \geq 5$	none.

(A.2)

Indeed, for $s = 1$ and $s = 2$ the constant terms are respectively $-4T - S + 2 < 0$ and $-3S + 3 < 0$. For $s = 3$ the constant term is $4pr - 5p - 5r + 4$: it is negative when one parameter equals 1, vanishes only at $(p, r) = (2, 2)$ among $p, r \geq 2$, and is positive otherwise. For $s = 4$ it is $8pr - 7p - 7r + 5$, which is negative at $(1, 1)$, zero at $(1, 2)$ and $(2, 1)$, and positive elsewhere. For $s = 5$ it equals $12pr - 9p - 9r + 6$; this vanishes only at $(1, 1)$ and is positive otherwise. Finally, increasing s by one increases the constant term by $4T - 2S + 1 = 4pr - 2p - 2r + 1 > 0$. Hence it is strictly positive for all $s > 5$.

Let $\lambda_1, \ldots, \lambda_5$ be the roots of $\Psi_{S,T,s}$ and define the pair-sum polynomial

$$\Pi_\Psi(y) = \prod_{1 \leq i < j \leq 5} \bigl(y - (\lambda_i + \lambda_j)\bigr).$$

It is the characteristic polynomial of the second additive compound of any matrix with characteristic polynomial Ψ. Since Ψ is the characteristic polynomial of a real symmetric quotient, all its roots are real.

We first calculate the polynomial whose roots are the pairwise sums.

Lemma A.1. *Let*

$$f(x) = x^5 + c_4 x^4 + c_3 x^3 + c_2 x^2 + c_1 x + c_0$$

have roots $\lambda_1, \ldots, \lambda_5$, counted with multiplicity, and write

$$\Pi_f(y) = \prod_{i<j} \bigl(y - (\lambda_i + \lambda_j)\bigr) = y^{10} + d_1 y^9 + \cdots + d_{10}.$$

Then

$$\begin{aligned}
d_1 &= 4c_4, \\
d_2 &= 3c_3 + 6c_4^2, \\
d_3 &= c_2 + 9c_3c_4 + 4c_4^3, \\
d_4 &= -3c_1 + 4c_2c_4 + 3c_3^2 + 9c_3c_4^2 + c_4^4, \\
d_5 &= -11c_0 - 5c_1c_4 + 2c_2c_3 + 5c_2c_4^2 + 6c_3^2c_4 + 3c_3c_4^3, \\
d_6 &= -22c_0c_4 - 2c_1c_3 - 2c_1c_4^2 - c_2^2 + 6c_2c_3c_4 + 2c_2c_4^3 + c_3^3 + 3c_3^2c_4^2, \\
d_7 &= -4c_0c_3 - 16c_0c_4^2 - 4c_1c_2 + c_2c_3^2 + 4c_2c_3c_4^2 + c_3^3c_4, \\
d_8 &= 7c_0c_2 - 9c_0c_3c_4 - 4c_0c_4^3 - 4c_1^2 - 3c_1c_2c_4 + c_1c_3^2 + c_1c_3c_4^2 \\
&\qquad - c_2^2c_3 + c_2^2c_4^2 + 2c_2c_3^2c_4, \\
d_9 &= 4c_0c_1 + 4c_0c_2c_4 - c_0c_3^2 - 4c_0c_3c_4^2 - 4c_1^2c_4 + c_1c_3^2c_4 - c_2^3 + c_2^2c_3c_4, \\
d_{10} &= -c_0^2 + 2c_0c_1c_4 + c_0c_2c_3 - c_0c_3^2c_4 - c_1^2c_4^2 - c_1c_2^2 + c_1c_2c_3c_4.
\end{aligned}$$

Proof. Put $u_m = \sum_{i=1}^{5} \lambda_i^m$ and $u_0 = 5$. With $a_1 = c_4, a_2 = c_3, a_3 = c_2, a_4 = c_1, a_5 = c_0$, Newton's identities give

$$\begin{aligned}
u_m + a_1 u_{m-1} + \cdots + a_{m-1} u_1 + m\, a_m &= 0 \qquad && (1 \leq m \leq 5), \\
u_m + a_1 u_{m-1} + \cdots + a_5 u_{m-5} &= 0 \qquad && (m > 5).
\end{aligned}$$

For the ten pair sums, their mth power sum is

$$v_m := \sum_{i<j} (\lambda_i + \lambda_j)^m = \frac{1}{2} \left(\sum_{r=0}^{m} \binom{m}{r} u_r u_{m-r} - 2^m u_m \right). \tag{A.3}$$

Indeed, the double sum over all ordered pairs (i, j) equals the binomial sum in parentheses before subtraction; the diagonal terms contribute $2^m u_m$, and every unordered pair with $i \neq j$ is counted twice. Finally Newton's identities for the degree-ten polynomial Π_f read

$$v_m + d_1 v_{m-1} + \cdots + d_{m-1} v_1 + m\, d_m = 0 \qquad (1 \leq m \leq 10). \tag{A.4}$$

Starting with $m = 1$ and applying (A.3)–(A.4) successively gives the displayed coefficients. For example, $v_1 = 4u_1 = -4c_4$, so $d_1 = 4c_4$; the case $m = 2$ gives $d_2 = 3c_3 + 6c_4^2$. The remaining eight lines follow by the same recurrence, with no root extraction or approximation. □

The following recurrence performs the polynomial reductions used below.

Lemma A.2. *Let*

$$\phi(t) = t^3 + At^2 + Bt + C, \qquad F(y) = y^{10} + d_1 y^9 + \cdots + d_{10},$$

and let α *satisfy* $\phi(\alpha) = 0$*. Set* $(u_0, v_0, w_0) = (0, 0, 1)$*.*

For $\theta = \beta - \alpha$*, define recursively, for* $1 \leq j \leq 10$*,*

$$\begin{aligned} u_j &= (A + \beta) u_{j-1} - v_{j-1}, \\ v_j &= B u_{j-1} + \beta v_{j-1} - w_{j-1}, \\ w_j &= C u_{j-1} + \beta w_{j-1} + d_j. \end{aligned} \tag{A.5}$$

Then

$$F(\beta - \alpha) = u_{10}\alpha^2 + v_{10}\alpha + w_{10}.$$

For $\theta = \beta + \alpha$*, the corresponding recurrence is*

$$\begin{aligned} u_j &= (\beta - A) u_{j-1} + v_{j-1}, \\ v_j &= -B u_{j-1} + \beta v_{j-1} + w_{j-1}, \\ w_j &= -C u_{j-1} + \beta w_{j-1} + d_j, \end{aligned} \tag{A.6}$$

and again $F(\beta + \alpha) = u_{10}\alpha^2 + v_{10}\alpha + w_{10}$*.*

Proof. Use Horner's rule. If the current remainder is $u\alpha^2 + v\alpha + w$, then

$$(\beta - \alpha)(u\alpha^2 + v\alpha + w) + d = -u\alpha^3 + (\beta u - v)\alpha^2 + (\beta v - w)\alpha + \beta w + d.$$

Since $\alpha^3 = -A\alpha^2 - B\alpha - C$, reduction modulo ϕ gives (A.5). Replacing $\beta - \alpha$ by $\beta + \alpha$ gives (A.6). To make the iteration transparent, the first two triples in the minus case are

$$(u_1, v_1, w_1) = (0, -1, \beta + d_1),$$

$$(u_2, v_2, w_2) = (1, -2\beta - d_1, \beta^2 + \beta d_1 + d_2),$$

and the remaining triples are obtained by the same three displayed linear updates. Induction on the Horner stage proves the assertion after ten steps. □

For $f = \Psi_{S,T,s}$, the five coefficients $c_4, \ldots, c_0$ are read from (6.4). Substitution into Lemma A.1, followed by Lemma A.2, gives the following terminal remainders. To make this finite calculation independently auditable, Supplementary File S1 lists, for each of the three rows below, the specialized coefficients $d_1, \ldots, d_{10}$ and every intermediate triple (u_j, v_j, w_j) for $1 \leq j \leq 10$. In particular, each entry in the terminal table is the last line of a displayed ten-step Horner calculation, not an unrecorded symbolic simplification.

$$
\begin{array}{c|c|c|c|c}
s & \text{argument} & u_{10} & v_{10} & w_{10} \\
\hline
1 & -3S-\alpha & -16C_2 & -16C_1 & 4C_2+8C_1-16H \\
2 & -(3S+1)-\alpha & -16A_2 & -16A_1 & -16A_0 \\
3,\ p=1 & 6+\alpha & -q_2(r) & -q_1(r) & -q_0(r)
\end{array}
\tag{A.7}
$$

Here the polynomials $C_2, C_1, H, A_2, A_1, A_0, q_2, q_1, q_0$ are displayed in the three cases below. Thus every pair-sum identity used later is obtained by the two explicit finite recurrences (A.4) and (A.5) or (A.6).

For later use, evaluation of (6.4) at $0, 1, 3, 5$ gives

$$
\begin{aligned}
&\Psi_{S,T,1}(0) = -21S - 20T + 54 < 0, && \Psi_{S,T,1}(1) = 16(S-1) > 0,\\
&\Psi_{S,T,1}(3) = -80T < 0, && \Psi_{S,T,1}(5) = 16(9S+19) > 0,\\
&\Psi_{S,T,2}(0) = -3(5S-9) < 0, && \Psi_{S,T,2}(1) = 32(S-1) > 0,\\
&\Psi_{S,T,2}(3) = -96T < 0, && \Psi_{S,T,2}(5) = 32(5S+11) > 0.
\end{aligned}
$$

The sign changes locate three distinct positive roots, one in each of $(0,1)$, $(1,3)$, and $(3,5)$. Since $\Psi(0) < 0$ and Ψ is monic of odd degree, the product of all five roots is $-\Psi(0) > 0$. The two remaining roots therefore have positive product. They cannot both be positive, because the total root sum is

$$-(3S + s - 7) = 7 - 3S - s \le 0,$$

whereas five positive roots would have positive sum. Thus, for $s = 1, 2$, Ψ has exactly three positive and two negative roots.

The case $s = 1$

Let α be the unique positive root of Φ. Since

$$\Phi(1/2) = -\frac{5}{8}(2S-1) < 0, \qquad \Phi(1) = 4T > 0,$$

we have

$$\frac{1}{2} < \alpha < 1. \tag{A.8}$$

Let B be the sum of the other two roots of Φ. Then $B = -3S - \alpha$.

Apply the first row of (A.7). Since $B = -3S - \alpha$, the terminal remainder is

$$
\begin{aligned}
\Pi_\Psi(B) &= -16C_2\alpha^2 - 16C_1\alpha + 4C_2 + 8C_1 - 16H\\
&= -16\left\{C_2\left(\alpha^2 - \frac{1}{4}\right) + C_1\left(\alpha - \frac{1}{2}\right) + H\right\}.
\end{aligned}
$$

Thus

$$\Pi_\Psi(B) = -16Q_1,$$

where

$$Q_1 = C_2\left(\alpha^2 - \frac{1}{4}\right) + C_1\left(\alpha - \frac{1}{2}\right) + H$$

and

$$
\begin{aligned}
C_2 &= 162S^5 + 891S^4 - 351S^3T + 1350S^3 - 1899S^2T + 152S^2 \\
&\quad - 3001ST - 920S + 144T^2 - 1821T + 669, \\
C_1 &= 486S^6 + 2754S^5 - 1053S^4T + 4374S^4 - 5994S^3T - 260S^3 \\
&\quad - 10012S^2T - 4166S^2 + 864ST^2 - 1678ST + 210S \\
&\quad + 2304T^2 + 2993T + 1210, \\
H &= 162S^6 + 1296S^5T + \frac{459}{2}S^5 + \frac{12825}{2}S^4T - \frac{13257}{4}S^4 \\
&\quad - 2808S^3T^2 + \frac{39969}{4}S^3T - \frac{20341}{2}S^3 - 13572S^2T^2 \\
&\quad + \frac{44985}{4}S^2T - 7664S^2 - 19968ST^2 + \frac{82135}{4}ST + 81S \\
&\quad - 8112T^2 + \frac{59837}{4}T - \frac{3055}{4}.
\end{aligned}
$$

We now verify $C_2, C_1, H > 0$ on (A.1). First,

$$
\frac{\partial C_2}{\partial T} = -351S^3 - 1899S^2 - 3001S + 288T - 1821.
$$

Using $T \leq S^2/4$ gives the explicit upper bound

$$
\frac{\partial C_2}{\partial T} \leq -351S^3 - 1827S^2 - 3001S - 1821 < 0.
$$

Similarly,

$$
\begin{aligned}
\frac{\partial C_1}{\partial T} = &\ -1053S^4 - 5994S^3 - 10012S^2 + 1728ST \\
&\ - 1678S + 4608T + 2993 \\
\leq &\ -1053S^4 - 5562S^3 - 8860S^2 - 1678S + 2993.
\end{aligned}
$$

For $S = z + 2$, the last line equals

$$
-1053z^4 - 13986z^3 - 67504z^2 - 137558z - 97147 < 0.
$$

At $T = S^2/4$, writing $S = z + 2$, one obtains

$$
\begin{aligned}
C_2 &= \frac{297}{4}z^5 + \frac{4671}{4}z^4 + \frac{27887}{4}z^3 + \frac{77765}{4}z^2 + 24612z + 11594, \\
C_1 &= \frac{891}{4}z^6 + \frac{7965}{2}z^5 + 28475z^4 + \frac{206921}{2}z^3 + \frac{796341}{4}z^2 \\
&\quad + 190393z + 70923.
\end{aligned}
$$

Thus $C_2, C_1 > 0$.

The coefficient of T^2 in H is

$$
-2808S^3 - 13572S^2 - 19968S - 8112 < 0,
$$

so H is concave as a function of T. It therefore suffices to check the endpoints of (A.1). With $S = z + 2$,

$$\begin{aligned} H(S, S-1) &= 1458z^6 + 20034z^5 + \frac{210339}{2}z^4 + \frac{537455}{2}z^3 \\ &\quad + \frac{721945}{2}z^2 + \frac{505125}{2}z + 52284, \\ H\left(S, \frac{S^2}{4}\right) &= \frac{297}{2}z^7 + \frac{23967}{8}z^6 + \frac{399297}{16}z^5 + \frac{1766055}{16}z^4 \\ &\quad + \frac{4414847}{16}z^3 + \frac{6077247}{16}z^2 + \frac{505125}{2}z + 52284. \end{aligned}$$

Hence $H > 0$, and (A.8) gives $Q_1 > 0$.

Let C be the sum of the two negative roots of Ψ, and let P be the sum of its three positive roots. Because the remaining three roots are positive, C is the smallest of the ten pair sums appearing as the roots of Π_Ψ. Since $\Pi_\Psi(y) > 0$ for every $y < C$ and $\Pi_\Psi(B) < 0$, it follows that $C < B$. The trace identities for Φ and Ψ give

$$P + C = \alpha + B + 6.$$

Therefore $P > \alpha + 6$.

The case $s = 2$

Here $0 < \alpha < 1$ and $B = -(3S+1) - \alpha$. The second row of (A.7), with $B = -(3S+1) - \alpha$, gives directly

$$\Pi_\Psi(B) = -16(A_2\alpha^2 + A_1\alpha + A_0), \tag{A.9}$$

where

$$\begin{aligned} A_2 &= 162S^5 + 945S^4 - 378S^3T + 1800S^3 - 2340S^2T + 1684S^2 \\ &\quad - 4896ST + 1088S + 320T^2 - 5760T + 4128, \\ A_1 &= 486S^6 + 2997S^5 - 1134S^4T + 6372S^4 - 7668S^3T + 5484S^3 \\ &\quad - 18840S^2T + 4864S^2 + 1920ST^2 - 18496ST + 10496S \\ &\quad + 6528T^2 - 13408T + 9664, \\ A_0 &= 1296S^5T - 567S^5 + 6750S^4T - 5553S^4 - 3024S^3T^2 \\ &\quad + 11142S^3T - 14772S^3 - 15120S^2T^2 + 15852S^2T - 12164S^2 \\ &\quad - 22848ST^2 + 33440ST - 3008S - 9408T^2 + 27872T - 9696. \end{aligned}$$

The derivatives satisfy

$$\begin{aligned} \frac{\partial A_2}{\partial T} &= -2(189S^3 + 1170S^2 + 2448S - 320T + 2880), \\ \frac{\partial A_1}{\partial T} &= -2(567S^4 + 3834S^3 + 9420S^2 - 1920ST \\ &\qquad\qquad + 9248S - 6528T + 6704). \end{aligned}$$

The upper bound $T \leq S^2/4$ gives, respectively,

$$189S^3 + 1170S^2 + 2448S - 320T + 2880 \geq 189S^3 + 1090S^2 + 2448S + 2880 > 0,$$
$$\begin{aligned} &567S^4 + 3834S^3 + 9420S^2 - 1920ST \\ &\quad + 9248S - 6528T + 6704 \\ &\geq 567S^4 + 3354S^3 + 7788S^2 + 9248S + 6704 > 0. \end{aligned}$$

Hence both derivatives are negative. At the upper endpoint, with $S = z + 2$,

$$\begin{aligned} A_2 &= \frac{135}{2}z^5 + 1055z^4 + 6316z^3 + 18220z^2 + 26536z + 20128, \\ A_1 &= \frac{405}{2}z^6 + 3630z^5 + 26220z^4 + 97820z^3 \\ &\quad + 200952z^2 + 227984z + 128064. \end{aligned}$$

Thus $A_2, A_1 > 0$.

The coefficient of T^2 in A_0 is

$$-3024S^3 - 15120S^2 - 22848S - 9408 < 0.$$

For $S \geq 3$, write $S = z + 3$. At the two endpoints,

$$\begin{aligned} A_0(S, S-1) &= 1296z^6 + 25191z^5 + 192672z^4 + 739020z^3 \\ &\quad + 1500414z^2 + 1532125z + 569410, \\ A_0\left(S, \frac{S^2}{4}\right) &= 135z^7 + \frac{7155}{2}z^6 + \frac{79341}{2}z^5 + 237492z^4 \\ &\quad + 822272z^3 + \frac{3247923}{2}z^2 + \frac{3298181}{2}z + 613593. \end{aligned}$$

Hence $A_0 > 0$ for $S \geq 3$. If $S = 2$, then $p = r = 1$ and $\alpha^2 + 6\alpha - 3 = 0$; in this case

$$A_2\alpha^2 + A_1\alpha + A_0 = 32(629\alpha^2 + 4002\alpha - 1017) = 32(870 + 228\alpha) > 0.$$

Thus the right-hand side of (A.9) is negative. The same pair-sum and trace argument as in the case $s = 1$ gives $P > \alpha + 6$.

The case $s = 3$

By (A.2), we may assume $p = 1$ and write $r \geq 1$. Evaluation gives

$$\Psi(1) = 48(S-1) > 0, \qquad \Psi(3) = -112T < 0, \qquad \Psi(5) = 16(11S + 25) > 0,$$

so Ψ has at least two positive roots. The third row of (A.7), obtained from the plus recurrence (A.6) with $\beta = 6$, gives

$$\Pi_\Psi(6 + \alpha) = -Q_3(r, \alpha),$$

where

$$\begin{aligned} Q_3 &= q_2(r)\alpha^2 + q_1(r)\alpha + q_0(r), \\ q_2(r) &= -28161r^4 + 135768r^3 - 290986r^2 + 642712r - 375173, \\ q_1(r) &= -65088r^4 + 742060r^3 - 940880r^2 + 3384020r - 2062336, \\ q_0(r) &= 85812r^4 + 445324r^3 + 861124r^2 + 241668r + 346008. \end{aligned}$$

For $r = 1, 2, 3$, the triples (q_2, q_1, q_0) are respectively

$$(84160, 1057776, 1979936), \quad (381875, 5837256, 9209424), \quad (318784, 14385296, 27795648),$$

so $Q_3 > 0$. Let now $r = d + 4$. Then

$$-q_2(r) = 28161d^4 + 314808d^3 + 1365226d^2 + 2377528d + 980165 > 0.$$

Moreover, $\Phi(0) = -(r + 1) < 0$ and

$$64\Phi(1/4) = 108r - 123 > 0,$$

so $0 < \alpha < 1/4$. Since Q_3 is concave in α, it suffices to check the endpoints. We have $Q_3(r, 0) = q_0(r) > 0$, and

$$\begin{aligned} 16Q_3\left(r, \frac{1}{4}\right) &= 1084479d^4 + 27580856d^3 + 236583766d^2 \\ &\quad + 864461144d + 1156964091 > 0. \end{aligned}$$

Therefore $\Pi_\Psi(6 + \alpha) < 0$. Since $\Pi_\Psi(y) > 0$ for sufficiently large y, some pair sum of roots exceeds $6 + \alpha$. The largest pair sum is the sum of the two largest roots, and Ψ has at least two positive roots; hence the sum P of all positive roots satisfies $P > 6 + \alpha$.

The case $s = 4$

The only possible parameters are $p = r = 1$. In this case

$$\Phi = (x + 1)(x^2 + 8x - 1), \qquad \Psi = (x^2 - 4x - 1)(x^3 + 7x^2 - 17x - 7).$$

The positive root of the reduced cubic is $\alpha = \sqrt{17} - 4$. The quadratic factor of Ψ has positive root $2 + \sqrt{5}$, and the cubic factor has a positive root in $(2, 3)$. Therefore

$$P > 4 + \sqrt{5} > 2 + \sqrt{17} = 6 + \alpha.$$

The middle inequality follows from $(2 + \sqrt{5})^2 = 9 + 4\sqrt{5} > 17$. This completes the proof of the root-sum inequality in all cases.

B Elementary inequalities for the asymmetric case

We prove Lemma 8.2. All estimates below use only the secular and normalization identities from Proposition 3.2, elementary rational inequalities, and positive-coefficient polynomial expansions. For each clearing of denominators we state the positive denominator and the exact resulting identity. Supplementary File S1 repeats these transformations in a single calculation and records the coefficient-by-coefficient cancellations in the two exceptional quadratic cases.

We shall use the following elementary facts without further comment. A polynomial in nonnegative variables whose coefficients are all nonnegative and whose constant term is positive is strictly positive. Also, if a quadratic q is concave on $[0, \Omega]$ and both endpoint values are positive, then $q > 0$ throughout the interval; if its leading coefficient is nonnegative and $q'(\Omega) < 0$, then q' is negative on $[0, \Omega]$, so q is decreasing and $q \geq q(\Omega) > 0$. These are the only positivity principles used below.

Let the non-singleton part sizes of G be $n_1, \ldots, n_\ell$, including the endpoint part sizes a and b, and write

$$L_i = \alpha + 2n_i - 3.$$

Combining (3.3) and (3.4) gives the useful identity

$$\frac{1}{\omega} = \frac{1 + \displaystyle\sum_{i=1}^{\ell} \frac{2n_i(n_i-1)}{L_i^2}}{1-\alpha}. \tag{B.1}$$

Indeed, (3.3) first gives

$$\frac{1}{\omega} = \frac{1 + \sum_i n_i/L_i}{1-\alpha} + \sum_i \frac{n_i}{L_i^2},$$

and each pair of summands combines as

$$\frac{n_i}{L_i(1-\alpha)} + \frac{n_i}{L_i^2} = \frac{n_i(L_i+1-\alpha)}{L_i^2(1-\alpha)} = \frac{2n_i(n_i-1)}{L_i^2(1-\alpha)}.$$

Since $0 < \alpha < 1$,

$$\frac{2n_i(n_i-1)}{L_i^2} > \frac{2n_i(n_i-1)}{(2n_i-2)^2} = \frac{n_i}{2(n_i-1)} > \frac{1}{2}.$$

The final asymmetric case contains at least three non-singleton parts, and therefore

$$0 < \omega < \frac{2}{5}. \tag{B.2}$$

The range $3 \le a < b$

Put

$$B_0 = \frac{a+b-2}{ab}.$$

Using (B.2), $L_a \ge 2a-3$, and $L_b \ge 2b-3$ in (8.15), we obtain

$$\mathcal{B} > B_0 - \frac{2(b-2)}{5b(2a-3)^2} - \frac{2(a-2)}{5a(2b-3)^2} - \frac{16}{25(2a-3)^2(2b-3)^2}.$$

Denote this lower bound by $B^-_{a,b}$. The denominator $25ab(2a-3)^2(2b-3)^2$ is positive. With $a = u+3$ and $b = v+3$, direct collection gives the exact identity

$$\begin{aligned} &25ab(2a-3)^2(2b-3)^2 \left(B^-_{a,b} - \frac{4}{5} B_0 \right) \\ &\quad = R_B(u,v), \end{aligned}$$

where

$$\begin{aligned} R_B(u,v) = {} & 80u^3v^2 + 200u^3v + 60u^3 + 80u^2v^3 + 800u^2v^2 \\ & + 1700u^2v + 780u^2 + 200uv^3 + 1700uv^2 + 3524uv \\ & + 1797u + 60v^3 + 780v^2 + 1797v + 936 > 0. \end{aligned}$$

Thus

$$\mathcal{B} > \frac{4}{5} B_0. \tag{B.3}$$

We next prove $\mathcal{G} < 6$. Since $a, b \ge 3$, we have $L_a \ge a$ and $L_b \ge b$. Moreover,

$$\frac{a\omega}{L_a^2} = \mu_a^2, \qquad \frac{b\omega}{L_b^2} = \mu_b^2.$$

The normalized eigenvector z has a nonzero coordinate on every quotient cell and, in the present case, has coordinates outside each endpoint part. Hence $0 < \mu_a^2, \mu_b^2 < 1$, and therefore

$$0 < 1 - \frac{a\omega}{L_a^2} < 1, \qquad 0 < 1 - \frac{b\omega}{L_b^2} < 1.$$

Dropping the negative term in (8.16) and using (B.2), the positive correction to $6 - 3/a - 3/b$ is less than

$$C_{a,b} := \frac{2}{5}\left(\frac{1}{a^2} + \frac{1}{b^2}\right) + \frac{12}{5ab} + \frac{32}{25ab}\left(\frac{1}{a^2} + \frac{1}{b^2}\right) + \frac{32}{5a^2b^2}.$$

Again set $a = u + 3$, $b = v + 3$. Since $25a^3b^3 > 0$, the required comparison follows from the exact identity

$$\begin{aligned}
&25a^3b^3\left(\frac{3}{a} + \frac{3}{b} - C_{a,b}\right) \\
&= 75u^3v^2 + 440u^3v + 645u^3 + 75u^2v^3 + 1290u^2v^2 \\
&\quad + 5625u^2v + 7258u^2 + 440uv^3 + 5625uv^2 + 21440uv \\
&\quad + 25383u + 645v^3 + 7258v^2 + 25383v + 27954 > 0.
\end{aligned}$$

Hence

$$\mathcal{G} < 6. \tag{B.4}$$

By (8.19), (B.3), (B.4), and (B.2),

$$\frac{\mathcal{N}}{16} > \frac{16}{25}B_0^2 - \frac{12}{5(2a-3)(2b-3)} = \frac{4P(a,b)}{25a^2b^2(2a-3)(2b-3)},$$

where

$$\begin{aligned}
P(a,b) = 16a^3b - 24a^3 + 17a^2b^2 - 136a^2b + 132a^2 \\
+ 16ab^3 - 136ab^2 + 328ab - 240a \\
- 24b^3 + 132b^2 - 240b + 144.
\end{aligned}$$

For $3 \le a < b$, this polynomial is positive except possibly at $(a,b) = (3,4)$ and $(3,5)$. The verification covers the parameter range in three disjoint pieces. If $a = 3$ and $b \ge 6$, write $b = t + 6$; if $a = 4$ and $b \ge 5$, write $b = t + 5$; and if $a \ge 5$, write $a = u + 5$, $b = u + v + 6$. Here $t, u, v \ge 0$, and expansion gives

$$\begin{aligned}
P(3, t+6) &= 24t^3 + 309t^2 + 1068t + 432, \\
P(4, t+5) &= 40t^3 + 460t^2 + 1520t + 860,
\end{aligned}$$

and

$$\begin{aligned}
P(u+5, u+v+6) = 49u^4 + 98u^3v + 758u^3 + 65u^2v^2 \\
+ 1120u^2v + 4197u^2 + 16uv^3 + 490uv^2 \\
+ 4074uv + 9540u + 56v^3 + 885v^2 \\
+ 4572v + 6912.
\end{aligned}$$

Every displayed coefficient is nonnegative and every constant term is positive. These three substitutions cover all $3 \le a < b$ except the two listed pairs.

The two exceptional pairs are handled by substituting into the preceding lower and upper bounds. For $(a, b) = (3, 4)$,

$$\mathcal{B} > \frac{2897}{7500}, \qquad \mathcal{G} < \frac{1525127}{337500}, \qquad \frac{\omega}{L_a L_b} < \frac{2}{75},$$

and hence

$$\frac{\mathcal{N}}{16} > \left(\frac{2897}{7500}\right)^2 - \frac{2}{75}\frac{1525127}{337500} = \frac{14528401}{506250000} > 0.$$

For $(a, b) = (3, 5)$,

$$\mathcal{B} > \frac{814}{2205}, \qquad \mathcal{G} < \frac{212782}{46305}, \qquad \frac{\omega}{L_a L_b} < \frac{2}{105},$$

so

$$\frac{\mathcal{N}}{16} > \left(\frac{814}{2205}\right)^2 - \frac{2}{105}\frac{212782}{46305} = \frac{237032}{4862025} > 0.$$

This proves $\mathcal{N} > 0$ whenever $3 \le a < b$.

The range $a = 2$

Here $L_a = 1 + \alpha$. Besides the 2-part and the b-part, at least one further non-singleton part is present. Therefore (B.1) gives

$$\frac{1}{\omega} > \frac{2 + 4/(1+\alpha)^2}{1-\alpha} > 5,$$

because

$$\frac{2 + 4/(1+\alpha)^2}{1-\alpha} - 5 = \frac{5\alpha^3 + 7\alpha^2 - \alpha + 1}{(1-\alpha)(1+\alpha)^2} > 0.$$

The denominator is positive, and the numerator is positive since $5\alpha^3 \ge 0$ and $7\alpha^2 - \alpha + 1 > 0$ (its discriminant is $1 - 28 < 0$). Thus

$$0 < \omega < \frac{1}{5}. \tag{B.5}$$

Assume first that $b \ge 5$. From (8.15) and (B.5),

$$\mathcal{B} > \frac{1}{2} - \frac{b-2}{5b} - \frac{4}{25(2b-3)^2}.$$

Dropping the negative term in (8.16) and bounding its remaining positive terms gives

$$\begin{aligned}\mathcal{G} <& \frac{9}{2} - \frac{3}{b} + \frac{1}{5}\left(1 + \frac{1}{(2b-3)^2}\right) + \frac{6}{5(2b-3)} \\ &+ \frac{8}{25(2b-3)}\left(1 + \frac{1}{(2b-3)^2}\right) + \frac{8}{5b(2b-3)}.\end{aligned}$$

Denote the displayed lower bound for $\mathcal{B}$ by B_b^- and the upper bound for $\mathcal{G}$ by G_b^+. For $b \ge 5$,

$$B_b^- > \frac{1}{2} - \frac{1}{5} - \frac{4}{25 \cdot 49} > 0, \qquad G_b^+ > 0,$$

and

$$\frac{\omega}{L_a L_b} < \frac{1}{5(2b-3)}.$$

It follows from (8.19) that

$$\frac{\mathcal{N}}{16} > (B_b^-)^2 - \frac{G_b^+}{5(2b-3)}.$$

The denominator $2500b^2(2b-3)^4$ is positive, and ordinary collection gives the exact identity

$$\begin{aligned}(B_b^-)^2 &- \frac{G_b^+}{5(2b-3)} \\ &= \frac{3600b^6 - 30800b^5 + 90000b^4 - 130980b^3 + 124119b^2 - 88380b + 32400}{2500b^2(2b-3)^4}.\end{aligned}$$

Consequently,

$$\frac{\mathcal{N}}{16} > \frac{3600b^6 - 30800b^5 + 90000b^4 - 130980b^3 + 124119b^2 - 88380b + 32400}{2500b^2(2b-3)^4}.$$

Writing $b = t + 5$, the numerator becomes

$$\begin{aligned}3600t^6 &+ 77200t^5 + 670000t^4 + 2969020t^3 \\ &+ 6909419t^2 + 7579310t + 2570975 > 0.\end{aligned}$$

Thus $\mathcal{N} > 0$ for $a = 2$, $b \geq 5$.

It remains to treat $b = 3, 4$. We now perform the substitution without suppressing the cancellation. Put $A = 1 + \alpha$ and $B = \alpha + 2b - 3$. Since $a = 2$, formula (8.15) becomes

$$\mathcal{B} = \frac{1}{2} - \frac{(b-2)\omega}{bA^2} - \frac{4\omega^2}{A^2B^2}.$$

Insert this expression and (8.16) into $\mathcal{N} = 16(\mathcal{B}^2 - \omega\mathcal{G}/(AB))$. The cancellation of the cubic and quartic terms can be checked before any further simplification. For $b = 3$ use the common denominator $9A^4B^4$, and for $b = 4$ use A^4B^4. In the following table, $N_k^{(B)}$ and $N_k^{(G)}$ are the coefficients of ω^k contributed by $16\mathcal{B}^2$ and by $16\omega\mathcal{G}/(AB)$, respectively, after multiplication by the stated common denominator:

(b,k)	$N_k^{(B)}$	$N_k^{(G)}$	$N_k^{(B)} - N_k^{(G)}$
$(3,3)$	$384B^2$	$384B^2$	0
$(3,4)$	2304	2304	0
$(4,3)$	$64B^2$	$64B^2$	0
$(4,4)$	256	256	0

The coefficients of degrees $0, 1, 2$, also recorded in Supplementary File S1, then give the following two explicit quadratics.

For $b = 3$, so that $B = \alpha + 3$,

$$\mathcal{N}_{2,3}(\omega) = \frac{36A^4B^4 - 24A^2B^3(23\alpha + 27)\omega - 16BR_3(\alpha)\omega^2}{9A^4B^4}, \tag{B.6}$$

where

$$R_3(x) = 18x^4 + 143x^3 + 423x^2 + 441x + 135.$$

For $b = 4$, so that $B = \alpha + 5$,

$$\mathcal{N}_{2,4}(\omega) = \frac{4A^4B^4 - 4A^2B^3(17\alpha + 25)\omega - 4BR_4(\alpha)\omega^2}{A^4B^4}, \tag{B.7}$$

where

$$R_4(x) = 8x^4 + 79x^3 + 305x^2 + 293x - 5.$$

Equations (B.6) and (B.7) are obtained by ordinary multiplication of the two displayed formulas for $\mathcal{B}$ and $\mathcal{G}$; in particular, they exhibit explicitly that no term of degree greater than two remains.

We next calculate the right endpoint in (8.22). Put $A = 1 + \alpha$ as above. Clearing the positive denominator in $\Omega_{2,b}^{-1}$ gives

$$2(1-\alpha)A^2(\alpha+3)^2\Omega_{2,3}^{-1} = D_3(\alpha), \qquad 2(1-\alpha)A^2(\alpha+5)^2\Omega_{2,4}^{-1} = D_4(\alpha).$$

Equivalently,

$$\Omega_{2,3}(\alpha) = \frac{2(1-\alpha)(\alpha+1)^2(\alpha+3)^2}{D_3(\alpha)}, \tag{B.8}$$

$$\Omega_{2,4}(\alpha) = \frac{2(1-\alpha)(\alpha+1)^2(\alpha+5)^2}{D_4(\alpha)}, \tag{B.9}$$

where

$$D_3(x) = 3x^4 + 24x^3 + 98x^2 + 168x + 123,$$
$$D_4(x) = 3x^4 + 36x^3 + 194x^2 + 356x + 323.$$

All denominators in (B.6)–(B.9) are strictly positive for $0 < \alpha < 1$.

Substitution of (B.8) into (B.6), followed by cancellation of the positive factor $(\alpha+1)^4(\alpha+3)^4$, yields

$$\mathcal{N}_{2,3}(\Omega_{2,3}) = \frac{4P_3(\alpha)}{9D_3(\alpha)^2}, \tag{B.10}$$

where

$$P_3(x) = 81x^8 + 1836x^7 + 17152x^6 + 91324x^5 + 287386x^4 + 507396x^3 + 466056x^2 + 176148x + 10125.$$

The coefficient of ω^2 in (B.6) is negative because $R_3(\alpha) > 0$. Moreover, $\mathcal{N}_{2,3}(0) = 4$ and the right-hand side of (B.10) is positive. The quadratic is therefore concave and positive at both endpoints, so it is positive throughout $[0, \Omega_{2,3}]$.

For $b = 4$, substitution of (B.9) into (B.7) gives

$$\mathcal{N}_{2,4}(\Omega_{2,4}) = \frac{4P_4(\alpha)}{D_4(\alpha)^2}, \tag{B.11}$$

where

$$P_4(x) = 9x^8 + 286x^7 + 3830x^6 + 27606x^5 + 111576x^4 + 241306x^3 + 278826x^2 + 144626x + 23679.$$

Differentiating (B.7) and substituting (B.9) gives

$$-\mathcal{N}'_{2,4}(\Omega_{2,4}) = \frac{4Q_4(\alpha)}{(\alpha+1)^2(\alpha+5)D_4(\alpha)}, \tag{B.12}$$

where

$$Q_4(x) = 19x^5 + 403x^4 + 3294x^3 + 10950x^2 + 15583x + 8055.$$

Every coefficient of P_4 and Q_4 is positive, so (B.11) and (B.12) are strictly positive. Also $\mathcal{N}_{2,4}(0) = 4$. If the leading coefficient in (B.7) is negative, the quadratic is concave and its positive endpoint values prove positivity on the entire interval. If that coefficient is nonnegative, its derivative is nondecreasing; since $\mathcal{N}'_{2,4}(\Omega_{2,4}) < 0$, the derivative is negative throughout $[0, \Omega_{2,4}]$. The quadratic is then decreasing and its minimum is the positive value at $\Omega_{2,4}$. Hence $\mathcal{N}_{2,4}(\omega) > 0$ for $0 < \omega < \Omega_{2,4}$.

All cases of Lemma 8.2 are now proved.

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